

Energy savings insurance implementation roadmap of India



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Foreword

India has long prioritised energy efficiency to strengthen energy security, improve productivity, and expand access to affordable energy for all. Key initiatives such as the Perform, Achieve and Trade scheme, Building Energy Codes, and the Ujala programme have helped drive progress across sectors.

In 2025, India was among the global leaders in energy efficiency, with more than 4% improvement in primary energy intensity. This aligns with the pace needed to meet global efficiency goals by 2030 and supports India's updated NDC, which targets a 47% reduction in GDP emission intensity by 2035 and 60% non-fossil installed power capacity by 2035. India has already achieved a 37.38% reduction in emission intensity by 2022 and 53.21% non-fossil installed power capacity by March 2026, underscoring the central role of energy efficiency in its clean energy transition.

The Bureau of Energy Efficiency is strengthening the energy efficiency market through innovative financing mechanisms. Its ADEETIE scheme is helping MSMEs adopt efficient technologies through interest subvention and technical support, while also improving project bankability.

Financial de-risking instruments such as energy savings insurance (ESI) have emerged as a promising tool to unlock energy efficiency investment. With a view to de-risk the financing for energy efficiency, BEE has been working with the OECD CEFIM Programme, to develop a roadmap for the design and implementation of ESI in India.

I believe this roadmap will guide stakeholders in designing and implementing an insurance-backed mechanism to support financing, unlock investment, accelerate deployment and support India's broader energy and climate goals.

Krushna Chandra Panigrahy

Director General, Bureau of Energy Efficiency (BEE)

Government of India



Editorial

Energy efficiency is a cornerstone of sustainable economic development. In India, it plays a pivotal role in reducing the energy intensity of a rapidly expanding economy, while enhancing affordability and energy security, boosting industrial competitiveness and mitigating emissions. Improving the efficiency of buildings, industry and transport is one of the fastest and most cost-effective ways to reduce energy demand and exposure to external shocks, while enhancing energy sovereignty and economic resilience. Over the past decades, India has made remarkable progress in strengthening its policy framework for energy efficiency. Numerous flagship initiatives led by the Indian Bureau of Energy Efficiency have progressively raised efficiency standards across end-use sectors. Micro, small and medium-sized enterprises, which form the backbone of India's economy, possess untapped potential for achieving energy savings, however barriers prevent higher investment in efficiency improvements.

In this context, the launch of the *Energy Savings Insurance (ESI) Implementation Roadmap of India*, developed in close collaboration between the Indian Bureau of Energy Efficiency of the Ministry of Power and the Organisation for Economic Co-operation and Development, marks an important milestone in unlocking finance and investment for energy efficiency in India. But, of course, it does not stop there. Implementing the roadmap requires collaboration between a multitude of national and international actors in raising awareness, strengthening capacities, piloting and technically supporting the rollout of the ESI scheme.

The OECD, through its Clean Energy Finance and Investment Mobilisation Programme, has worked with the BEE to strengthen India's enabling environment for energy efficiency financing and overcome persisting barriers. This roadmap emerges from a sustained and inclusive collaborative process. The 2022 *Clean Energy Finance and Investment Roadmap of India* identified Energy Savings Insurance schemes as a key instrument, following extensive cross-ministerial and multi-stakeholder consultations. In the following, the BEE tasked the OECD with developing a dedicated needs assessment and ESI implementation roadmap in India. A jointly organised series of workshops and consultations in 2024-2025 and international lessons from ESI deployment, confirmed the potential of energy savings insurance to substantially reduce the perceived risks of lenders and investors, thereby unlocking private capital for energy efficiency in India's MSMEs and beyond. This roadmap provides a step-by-step guide to tapping into the energy efficiency potential in high priority sectors of the Indian economy through ESI.

We extend our appreciation to the BEE for its leadership and commitment to this initiative. We equally thank the Asian Development Bank, the Ministry of MSMEs, the Insurance Regulatory and Development Authority of India, the International Institute for Energy Conservation, the many financial institutions, insurers, energy service companies, energy auditors and other experts who contributed to this work. We look forward to continuing our excellent collaboration with the BEE and all partners in supporting the mobilisation of clean energy finance and investment as we move from roadmap to implementation.

H.E. Jaime de Bourbon de Parme

Director Environment

Organisation for Economic Co-operation and Development (OECD)

A handwritten signature in black ink, reading "Jaime de Bourbon de Parme". The signature is written in a cursive style with a long, sweeping underline.

Acknowledgements

The Energy Savings Insurance (ESI) Implementation Roadmap of India has been developed by the Organisation for Economic Co-operation (OECD) in close collaboration with the Bureau of Energy Efficiency (BEE) through its Clean Energy Finance and Investment Mobilisation (CEFIM) programme under their joint work programme. Substantial inputs were provided by the International Institute for Energy Conservation (IIEC) and the IIEC Private Limited.

Building on the fruitful collaboration between the Government of India and the OECD, this roadmap supports India's efforts to realise its clean energy transition by providing guidance on the proposed implementation structure and critical steps to design and implement an energy savings insurance (ESI) scheme in India.

This roadmap is authored by Sanjay Dube (IIEC) and Cornelia Schenk (OECD CEFIM), with valuable inputs from Nupur Ahuja (BEE), and with guidance from Shri Palraj Shyam Sunder, Director (BEE) and Geraldine Ang, Energy Team Lead of the Clean Energy Finance and Investment Mobilisation (CEFIM) Programme, in the Finance, Investment and Global Relations Division of the OECD Environment Directorate. The team would like to thank Yuval Laster, Head of the Finance, Investment and Global Relations Division at the OECD Environment Directorate for his managerial guidance and support. The work was conducted under the leadership of Shri Krushna Chandra Panigrahy, Director General of the Bureau of Energy Efficiency (DG-BEE) and H.E. Jaime de Bourbon de Parme, Director, Environment Directorate of the OECD. The OECD CEFIM programme's work is being undertaken with funding from the Governments of Denmark and Canada.

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Abbreviations and acronyms

AASI	Indonesia Sharia Insurance Association
ADB	Asian Development Bank
ADEETIE	Assistance for Deployment of Energy Efficient Technologies in Industrial Establishments
AHU	Air handling unit
AMI	Advanced Metering Infrastructure
APFC	Automatic Power Factor Control
AXA	AXA Assurance Maroc
BEE	Bureau of Energy Efficiency
CESCE	Compañía Española de Seguros de Crédito a la Exportación
CEFIM	Clean Energy Finance and Investment Mobilisation
CO	Carbon monoxide
CRES	Centre for Renewable Energy Sources and Saving
DSCR	Debt Service Coverage Ratio
EEI	Energy efficiency insurance
ESI	Energy savings insurance
ESCO	Energy Service Company
GIC	General Insurance Council
HBOR	Croatian Bank for Reconstruction and Development
HVAC	Heating, ventilation and air conditioning
IEA	International Energy Agency

IE3	Premium efficiency class of electric motors
IIED	International Institute for Energy Conservation
IPMVP	International Performance Measurement and Verification Protocol
IRDAI	Insurance Regulatory and Development Authority of India
IRR	Internal rate of return
ISO	International Organization for Standardization
kWh	Kilowatt-hour
MSME	Micro, Small and Medium Enterprises
NAPCC	National Action Plan on Climate Change
NDC	Nationally Determined Contributions
NMEEE	National Mission for Enhanced Energy Efficiency
NPV	Net present value
OECD	Organisation for Economic Co-operation and Development
PSU	Public sector undertaking
SIE	Société d'Ingénierie Énergétique
SIDBI	Small Industries Development Bank of India
SLM	Straight Line Method (depreciation)
VFD	Variable frequency drive
VVR	Variable volume ratio
WB	World Bank
WHR	Waste heat recovery

Executive summary

India is at a critical juncture where rising energy demand needs to be met in alignment with its ambitious climate objectives and intentions to improve the efficiency of its economy. Increasing India's share of clean energy in total installed capacity and reducing the country's emissions intensity are significant steps towards achieving its sustainable development and climate commitments. As of October 2024, India installed 203 GW of renewable energy and reduced the emission intensity of its GDP by 33% (from 2005-2019), bringing it closer to achieving its 2030 Nationally Determined Contribution (NDC) targets. As India advances towards its goal of attaining developed nation status by 2047 under its "Viksit Bharat 2047" initiative, it remains committed to provide clean, reliable and affordable energy for all citizens.

India's energy sector is experiencing tremendous growth mirroring the country's booming population and economic development. Over the past decade, India's peak electricity demand has surged by 79%, while total electricity consumption doubled, growing at an annual rate of 5.8% (Bureau of Energy Efficiency, 2024^[1]).

India's energy policy reforms of the past twenty years focused on creating the enabling conditions to deliver on the government's ambitions to ensure affordable, secure and sustainable energy services for all Indians. The country's transition towards clean energy is a crucial component of India's overall energy strategy, which requires a holistic approach encompassing advances in technology, policy and investment. Policies promoting a tripling of renewable energy generation capacity and doubling of the rate of energy efficiency (EE) improvement by 2030 play a crucial role in accelerating this transition. Core policies promoting energy efficiency improvements in India include the 2008 National Action Plan on Climate Change (NAPCC), the 2011 National Mission for Enhanced Energy Efficiency (NMEEE) including the Perform, Achieve and Trade (PAT) scheme for industries, but also standards and labelling schemes for appliances, Unnat Jyoti by Affordable LEDs for All (UJALA) scheme, the Energy Conservation Building Code, Electric Vehicle Mission and demand side management programmes. In fiscal year 2023-24, these achieved energy savings of approximately 54 million tonnes of oil equivalent, which corresponds to 6% of the nation's total primary energy supply (Bureau of Energy Efficiency, 2024^[1]). Over the past twenty years, these have contributed to India's remarkable progress towards higher energy efficiency, with total energy supply per unit of GDP (PPP) decreasing by 36% between 2000 and 2023 (IEA, 2026^[2]).

Achieving India's ambitious clean energy and climate goals, however, requires the mobilisation of significant investment in energy efficiency improvements, which has to be supported through an enabling policy framework and appropriate financing and risk mitigation instruments to unlock the necessary capital.

Barriers to energy efficiency financing

Despite its potential, India's energy efficiency market faces several challenges that complicate the acceleration of energy efficiency improvements and scaling up of investments. Information asymmetries, capacity constraints, lack of access to capital and competing investment priorities remain key barriers to scaling the energy efficiency market in India, which are particularly challenging for MSMEs.

Many financiers, investors and end users lack a full understanding of the financial, environmental and operational benefits of energy efficiency projects, which is partly due to the unavailability of long-term performance data for energy efficient appliances and equipment available in the Indian market and leads to higher risk perception. While some energy efficiency measures such as energy management systems can be implemented at no or low costs, energy efficient technologies often require higher upfront investment compared to less efficient alternatives. In a price sensitive market like India with high-risk perception of customers regarding the future performance of efficient technologies and systems, and a short-term investment perspective, this can discourage investment in solutions that deliver significant reductions in operating expenses in the long run. According to the BEE, only around 130 ESCOs are currently empanelled¹. While the energy efficiency market in India is estimated to be worth INR 150 000 crore, Energy Service Companies (ESCOs) only tapped about 5% of this market potential (Bureau of Energy Efficiency, 2026^[3]).

These barriers can be addressed through a range of measures, including by strengthening the regulatory framework and local ESCO market, enhancing capacities for energy efficiency improvements and financing and incentivising investment in energy efficiency. Dedicated financing instruments for energy efficiency projects in different end-use sectors and the deployment of risk mitigation instruments such as guarantee and insurance products can further promote energy efficiency investment.

Aim of the roadmap and its audience

This Energy Savings Insurance (ESI) Implementation Roadmap aims to provide a step-by-step guidance for the design and implementation of an Energy Savings Insurance Scheme in India. It shall inform and guide the Bureau of Energy Efficiency (BEE), the Ministry of Power (MOP), the Insurance and Regulatory and Development Authority (IRDA) of India, as well as a range of other key stakeholders, including domestic financial institutions such as banks, non-banking financial institutions and insurance companies, but also insurance associations, ESCOs, energy efficiency technology providers and vendors, industry associations, validation entities, multilateral development banks and international institutions active in India.

What is an Energy Savings Insurance (ESI) scheme?

The Energy Savings Insurance (ESI) model has been designed to mitigate the performance risks of energy efficiency projects, thereby enhancing the creditworthiness of the investors in energy efficient technologies and processes. The ESI scheme is typically implemented as a package of financial and non-financial components, including the energy savings insurance product, a standardised energy performance contract with guaranteed savings, independent technical validation and existing or new financial instruments and incentives that can be linked to energy efficiency projects backed by an ESI. These components work together to reduce perceived risks, build trust in future energy savings among involved stakeholders and mobilise enterprises to invest in energy efficiency solutions.

The model provides a performance guarantee that the contracted energy savings will be achieved through efficiency measures implemented in the project. If the project does not achieve the pledged savings, the provider is obliged to either make necessary improvements or to compensate the client. Should this fail, the guarantor, usually a specialised insurance agency, will financially compensate the client.

¹ The BEE rates energy service companies (ESCOs) in terms of their capacity and experience in implementing energy efficiency projects based on performance contracting, availability of technical manpower, financial strength etc. This rating is made publicly available to help potential public and private clients select an ESCO and facilitate energy efficiency project implementation.

The Energy Savings Insurance (ESI) scheme could therefore be instrumental in overcoming persisting market barriers and mobilising investment in energy efficiency in India. International experience in implementing the ESI model, for instance in Latin America, Europe and East Asia, demonstrates its potential to mobilise investment in energy efficiency improvements and achieve the committed savings, as well as its adaptability to specific national contexts and objectives. Key lessons learned from these examples have contributed to shaping the design of the proposed ESI implementation structure in India.

ESI market opportunity, structure and key actors

India's energy efficiency market holds immense potential for scale, driven by rapid industrialisation, urbanisation and increasing energy demand. According to estimates by the Bureau of Energy Efficiency (BEE), energy efficiency investments across key sectors could reach INR 1 002 329 crores to 1 320 630 crores (USD 121-160 billion) by 2031 (OECD, 2024^[6]).

The industrial MSME sector was identified as high-impact opportunity for the implementation of the ESI scheme, thanks to its potential for energy efficiency improvements and for mobilising investment. MSMEs account for 25% of India's energy consumption in the industry sector. Driven by the nation's economic growth and MSME growth, amongst other factors, this consumption is expected to increase by 50% by 2030. Leveraging a unique Indian characteristic, the implementation of an ESI scheme through MSME clusters appears as a promising approach to implement energy efficiency improvements, mobilise financing and achieve better economies of scale and lower transaction costs for participating actors. Here, MSMEs are the beneficiaries of the energy savings insurance, while a standardised energy performance contract is signed between the MSME cluster and an energy service company (ESCO). Following an initial rollout of the ESI scheme in industrial SMEs and at cluster level, the ESI scheme could be expanded to the building sector, and particularly public buildings. The building sector is currently responsible for approximately 35% of India's total energy consumption (Government of India, 2008^[4]).

The effective design and implementation of the ESI scheme in India requires close co-ordination and collaboration between a range of stakeholders, capitalising on their unique mandates and strengths. These include public and private sector agencies, financial institutions, validation entities, ESCOs, individual experts and energy auditors, IT and communications experts, regulatory bodies and international institutions. The implementation of an ESI scheme further benefits from technical and financial contributions of development partners, multilateral development banks (MDBs), regional development banks and other development finance institutions (DFIs). The proposed ESI implementation structure explains the roles of the BEE and other public authorities, ESI providers, financial institutions, ESCOs, independent validation entity, MSMEs and the mechanism for fund flow. Linking the ESI scheme with the ADEETIE scheme is a promising approach to enhance the impact of both schemes in promoting energy efficiency improvements in industrial MSMEs.

Implementation plan in India

The proposed implementation plan, or implementation roadmap, provides a step-by-step guide to core actors under the leadership of the BEE for the design and rollout of an effective ESI scheme.

Preparatory phase

As part of the preparatory efforts to introduce an ESI scheme in India, **the support from key stakeholders needs to be mobilised to ensure a sustainable anchoring in the Indian market and sufficient uptake of the insurance product.** Key domestic stakeholders for the ESI scheme include the General Insurance Council (GIC), participating banks under the ADEETIE scheme and other interested financial institutions,

insurance companies (to be identified with support from GIC), the Insurance Regulatory and Development Authority (IRDAI), ESCOs and energy efficiency technology providers. The implementation of an ESI scheme further benefits from technical and financial contributions of development partners, multilateral development banks (MDBs), regional development banks and other development finance institutions (DFIs).

Preparatory steps for the implementation of an ESI scheme in India include:

- Stakeholder engagement and consultations.
- Needs assessment and identification of priority sectors for implementation.
- Identification of priority MSME clusters and sub-sector for piloting the ESI scheme.
- Design of the ESI structure and definition of stakeholder roles.

Implementation phase

The implementation phase of the ESI scheme covers necessary steps to fully operationalise it, pilot it in selected MSME clusters and, based on the learnings derived from this process, roll out the scheme in India.

Key steps to operationalise the ESI scheme include:

- Determining a facilitating entity.
- Identifying and mandating an independent validation entity and defining monitoring and verification methodologies.
- Developing standardised energy performance contracts and document templates.
- Developing the energy savings insurance product and filing for approval.
- Designing capacity building and awareness raising campaigns.

Next steps

The BEE will launch the ESI Implementation Roadmap in alignment with the ADEETIE scheme in 2026 to operationalise the ESI scheme.

The BEE will convene a Technical Committee (TC) comprised by main actors for the implementation of the ESI scheme to coordinate implementation activities and ensure alignment with the ADEETIE scheme. The TC should then convene regular meetings under the chairmanship of the DG-BEE.

The design and implementation of ESI pilot projects, and the creation of a pipeline of projects could be coordinated by the TC with guidance from the BEE.

The BEE, supported by the facilitating entity and other partners, should (i) identify capacity building needs of key stakeholders and define targeted training programmes, and (ii) develop standardised templates, contracts and other documents.

Based on the learnings from an initial rollout of both the ESI scheme and the ADEETIE scheme, the BEE could develop a long-term strategy for embedding the ESI scheme in phase II of ADEETIE.

1 Introduction

Context and rationale

India is at a critical juncture where rising energy demand needs to be met in alignment with its ambitious climate objectives and intentions to improve the efficiency of its economy. Increasing India's share of clean energy in total installed capacity and reducing the country's emissions intensity are significant steps towards achieving its sustainable development and climate commitments. As of October 2024, India installed 203 GW of renewable energy and reduced the emission intensity of its GDP by 33% (from 2005-2019), bringing it closer to achieving its 2030 Nationally Determined Contribution (NDC) targets. As India advances towards its goal of attaining developed nation status by 2047 under its "Viksit Bharat 2047" initiative, it remains committed to provide clean, reliable and affordable energy for all citizens.

India's energy sector is experiencing tremendous growth mirroring the country's booming population and economic development. Total primary energy supply has increased by 54.5% over the past decade (2013-14 to 2023-24), rising from 589 Mtoe to 910 Mtoe. Over the same time frame, India's peak electricity demand has surged by 79%, rising from 136 GW to 243 GW. Notably, on 30 May 2024, India recorded an all-time high peak electricity demand of 250 GW. Total electricity consumption (including captive) doubled from 874 TWh to 1 543 TWh between 2013-14 and 2023-24, growing at an annual rate of 5.8% (Bureau of Energy Efficiency, 2024^[1]).

Over the past twenty years, India's energy policy reforms focused on creating enabling conditions to deliver on the government's ambitions to achieve an affordable, secure and sustainable energy system. While fossil fuels like coal and oil continue to dominate the energy mix, the government places growing emphasis on greening the energy mix through the deployment of renewable energy sources such as solar, wind and biomass along with battery energy storage and new technologies like green hydrogen and green ammonia. India's clean energy transition is a crucial component of India's overall energy strategy, which requires a holistic approach encompassing advances in technology, policy and investment. Policies promoting a tripling of renewable energy generation capacity and doubling of the rate of energy efficiency (EE) improvement by 2030, including through strengthened regulations, information and capacity building measures as well as incentives for technology innovation and deployment and investment play a crucial role in accelerating this transition. The role of accurate and timely energy data is instrumental in formulating informed policies to support energy transition.

India has shown remarkable progress towards energy efficiency improvements, with total energy supply per unit of GDP (PPP) decreasing by 36% between 2000 and 2023 (IEA, 2026^[2]). In 2025, India was among the world's leaders in energy efficiency, achieving over 4% improvement in primary energy intensity compared to the previous year (IEA, 2025^[5]), which corresponds to the target rate of global efficiency progress by 2030. Significant policy focus has been placed on energy efficiency improvements to ensure energy security, productivity and energy access for all its citizens. Key policies promoting energy efficiency improvements include the 2008 National Action Plan on Climate Change (NAPCC) (Government of India, 2008^[4]), the 2011 National Mission for Enhanced Energy Efficiency (NMEEE) (Government of India, 2011^[6]) including the Perform, Achieve and Trade (PAT) scheme for industries, but also standards and labelling schemes for appliances, Unnat Jyoti by Affordable LEDs for All (UJALA) scheme promoting the

deployment of LED technology, the Energy Conservation Building Code, Electric Vehicle Mission and demand side management programmes (Ministry of Power, 2024^[7]). The implementation of India's energy efficiency programmes and schemes resulted in cumulative energy savings of 53.60 million tonnes of oil equivalent (Mtoe) in 2023-24, and cost savings of INR 200 212.84 crores (USD 23.6 billion) (Bureau of Energy Efficiency, 2024^[11]).

To ensure the achievement of its ambitious clean energy and climate goals, including its net-zero emissions target by 2070, however, the country needs to mobilise significant investment in energy efficiency, including investments in end-use electrification, conservation and demand-side management. **Scaling up investment in energy efficiency improvements not only requires a supportive policy framework and appropriate financing instruments but also dedicated risk mitigation instruments to overcome the often high risk perception of energy efficiency projects and unlock the necessary capital.**

Financial de-risking for doubling energy efficiency

Despite its potential, India's energy efficiency market faces several challenges that complicate the acceleration of energy efficiency improvements and scaling up of investment. Information asymmetries, capacity constraints, lack of access to capital and competing investment priorities remain key barriers to scaling the energy efficiency market in India, and are particularly challenging for MSMEs. Many financiers, investors and end users lack a full understanding of the financial, environmental and operational benefits of energy efficiency projects which is partly due to the unavailability of long-term performance data for energy efficient appliances and equipment available in the Indian market and leads to higher risk perception. While some energy efficiency measures such as energy management systems can be implemented at no or low costs, energy efficient appliances and equipment often require higher upfront investment compared to less efficient alternatives. In a price sensitive market like India with high-risk perception of customers regarding the future performance of efficient technologies and systems, and a short-term investment perspective, this can discourage investment in solutions that deliver significant reductions in operating expenses in the long run. According to the BEE, only around 130 ESCOs are currently empanelled. While the energy efficiency market in India is estimated to be worth INR 150 000 crore, Energy Service Companies (ESCOs) only tapped about 5% of this market potential (Bureau of Energy Efficiency, 2026^[3]).

The **Energy Savings Insurance (ESI) scheme** could be instrumental in overcoming persisting market barriers and mobilising the necessary investments in energy efficiency. When integrated into the country's broader financial and policy framework, ESI can enhance investor confidence, stimulate financing from financial institutions and encourage technology providers and energy service companies (ESCOs) to scale up energy efficiency solutions. Lessons from other countries that have successfully implemented an ESI scheme or other financial risk mitigation mechanisms can further inform India's approach and support the expansion of energy efficiency investments at national level.

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This Energy Savings Insurance (ESI) Implementation Roadmap aims to provide a step-by-step guidance for the design and implementation of an Energy Savings Insurance scheme in India. It shall inform and guide the Bureau of Energy Efficiency (BEE), the Ministry of Power (MOP), the Insurance and Regulatory and Development Authority (IRDA) of India, as well as a range of other key stakeholders, including not only domestic financial institutions such as banks, non-banking financial institutions and insurance companies, but also insurance associations, ESCOs, energy efficiency technology providers and vendors,

industry associations, validation entities, multilateral development banks and international institutions active in India.

What is an Energy Saving Insurance (ESI) scheme?

An Energy Savings Insurance (ESI) scheme is an instrument designed to mitigate the performance risks of energy efficiency projects, thereby enhancing the creditworthiness of the investors in energy efficient technologies and processes. The ESI product covers the energy savings commitments made by the technology supplier or ESCO under an energy performance contract for a specified period of time. If the project does not achieve the pledged savings, the provider is obliged to either make necessary improvements or to compensate the client. By creating a financial liability for the technology provider tied to their performance commitment, ESI ensures accountability. If the provider fails to honour this contractual commitment, the guarantor, usually a specialised insurance agency, will financially compensate the client.

The ESI scheme is typically implemented as a package of financial and non-financial elements, including the ESI product, a standardised energy performance contract with guaranteed savings, independent technical validation and financing. These components work together to reduce perceived risks, build trust in future energy savings among involved stakeholders, and mobilise enterprises to invest in energy efficiency solutions. In detail, it comprises:

- A standardised, simplified energy performance contract which defines the energy savings commitment and the methodology how these are measured, validated, guaranteed and insured.
- An independent technical validation process through a credible third-party entity, which evaluates the project's savings potential, verifies the installation and acts as an arbitrator.
- An ESI product (often a surety bond), which insures the enterprise against the technology provider/ESO failing to fulfil their energy savings obligation.
- Existing or new financial instruments and incentives which can be linked to EE projects with an ESI.

ESI provides a performance guarantee that the contracted energy savings will be achieved through efficiency measures implemented in the project. This is based on a contractual relationship between the technology provider and their client, including performance guarantee coverage and independent technical validation. If the project obtains bank financing, a bank-borrower relationship is created that is independent of the client-provider relationship.

The ESI model has been tested and implemented in multiple countries, including in Latin America, Europe and Asia. These examples showcase its potential to mobilise investments and achieve energy savings, while also demonstrating its adaptability to the specific national context and objectives. Lessons from other countries that have successfully implemented an ESI scheme or other financial risk mitigation mechanisms can further inform India's approach and support the expansion of energy efficiency investments at the national level. The availability of robust data collection frameworks, real-time monitoring of energy savings, and enhanced collaboration between public and private stakeholders along with improved market intelligence are among the success factors for the implementation of the ESI scheme.

About this roadmap

Based on the learnings from ESI schemes in other markets, **this ESI implementation roadmap provides a step-by-step guide to its introduction in India to unlock investment in energy efficiency in high priority sectors of the economy**, thereby contributing to the achievement of India's sustainable

development goals. The roadmap further outlines the intended ESI implementation structure, financial modelling and international examples to guide the scheme's adaptation and scalability in India.

The Organisation for Economic Co-operation and Development (OECD), through its Clean Energy Finance and Investment Mobilisation (CEFIM) programme, assists emerging economies in mobilising private finance and investment for clean energy, including renewable energy, energy efficiency, transmission grids, clean hydrogen and industry transformation. It provides targeted, demand-driven support to improve domestic enabling conditions, unlock blended finance and other risk mitigation instruments, and strengthen domestic financial markets in line with the needs of each partner country. The programme supports partner countries including Colombia, Egypt, India, Indonesia, the Philippines, South Africa, Thailand and Viet Nam, as well as a broader set of emerging markets and developing economies (EMDEs).

In India, the OECD CEFIM programme aims to collaborate with key national stakeholders and provide recommendations to unlock and accelerate the finance and investment needed to meet the country's clean energy goals. Following close engagement between the OECD, the Government of India, the Natural Resources Defense Council (NRDC) and other clean energy stakeholders, the [Clean Energy Finance and Investment Roadmap of India](#) ("the CEFI Roadmap") was launched in October 2022 (OECD, 2022^[8]). The CEFI Roadmap provided non-prescriptive recommendations for the Government of India to **strengthen the domestic enabling environment to scale up finance and investment in (i) energy efficiency in micro, small and medium enterprises (MSMEs), as well as (ii) offshore wind power and (iii) green hydrogen production.** The CEFI Roadmap was developed in collaboration with and guided by cross-ministerial working groups and conducting multiple stakeholder consultations between 2021 and 2022, involving more than 50 national and international organisations. This process helped identify key barriers and opportunities, innovative financing solutions and investment mechanisms for unlocking investment in those three clean energy sectors. Based on the government's priorities and extensive stakeholder consultations, the establishment of an Energy Savings Insurance (ESI) scheme emerged as one of the CEFI Roadmap's key recommendations to accelerate energy efficiency improvements.

Building on the CEFI Roadmap recommendations, the Bureau of Energy Efficiency (BEE) of the Indian Ministry of Power identified the ESI scheme as a key initiative to boost energy efficiency financing, particularly for MSMEs. Consequently, the OECD CEFIM programme organised focused discussions and an international workshop on "[Energy Efficiency De-Risking Instruments in India: The Role of Energy Savings Insurance and Other Instruments](#)" in February 2024. Building on these discussions, the BEE tasked the OECD CEFIM programme with developing a needs assessment and implementation roadmap for the ESI scheme.

Developing an ESI implementation roadmap necessitates collaboration and backing from key stakeholders, such as the BEE under the Ministry of Power, the Ministry of MSMEs, the Insurance Regulatory and Development Authority of India (IRDAI), commercial banks, insurance companies and other financial institutions, large industries, energy efficiency solution providers/ESCOs and MSMEs. The implementation of an ESI scheme further benefits from technical and financial contributions of development partners, multilateral development banks (MDBs), regional development banks and other development finance institutions (DFIs). Following its mandate, the OECD CEFIM programme, in collaboration with the BEE and supported by the International Institute for Energy Conservation (IIEC), conducted a series of consultation workshops with ESCOs, MSMEs, energy efficiency technology providers and energy auditors in October 2024.

On 20th March 2025, the OECD in partnership with the Asian Development Bank (ADB) and the Bureau of Energy Efficiency (BEE) organised a joint roundtable consultation on "[De-Risking Instruments for Energy Efficiency Finance in India: The role of energy savings insurance and other instruments](#)" with financing institutions and insurance companies in Mumbai, India. The IIEC facilitated the delivery of the workshop and provided technical inputs. The suggested structure for the ESI scheme and a case for

the design and implementation of an ESI pilot was presented to the stakeholders for their inputs and suggestions, which were later incorporated in this roadmap.

Outline of the roadmap

Chapter 1 highlights the importance of energy efficiency for achieving India's energy and sustainable development objectives, explains the importance of de-risking investment in energy efficiency projects to mobilise additional capital and introduces the energy savings insurance (ESI) scheme.

Chapter 2 discusses the specific market opportunity and barriers to energy efficiency investment in India.

Chapter 3 highlights the potential of MSMEs to become first adopters of an ESI in India, with a particular view to its implementation at cluster level in the manufacturing sector.

Chapter 4 provides the actual ESI implementation plan, outlining critical steps for the preparatory phase and implementation phases, including proposed timelines for implementation.

Chapter 5 introduces additional considerations to facilitate the implementation of the ESI scheme in India, starting from the establishment of an energy consumption baseline, to the economic analysis of suitable energy efficient technologies, and the validation process and methodology.

Chapter 6 provides an exemplary assessment of financial viability and savings potentials of selected pilot applications,

Chapter 7 finally summarises key next steps for the operationalisation of the ESI implementation roadmap.

2 Unlocking energy efficiency investment in India

Unlocking the potential of India's energy efficiency market is critical to achieve energy affordability, economic competitiveness and poverty reduction, as well as climate goals. Over the past two decades, the country has made significant progress in expanding electricity access to over 700 million people and driving clean energy adoption, including solar power generation and energy-efficient technologies. Energy policy reforms have significantly contributed to achieving an affordable, secure and sustainable energy system, leading to remarkable growth in renewable energy capacity and energy efficiency improvement. Energy consumption per unit of GDP (energy intensity) in India has decreased over the past decade, indicating progress towards higher energy efficiency and sustainable development. The total primary energy supply per 1 000 INR of GDP declined from 0.006 toe/1 000 INRs in 2013-14 to 0.005 toe/1 000 INRs in 2023-24, suggesting enhanced energy efficiency and utilisation. Similarly, total final energy consumption per 1 000 INR of GDP decreased by 2% annually between 2014 and 2024 (Bureau of Energy Efficiency, 2024^[11]).

As one prominent example, the UJALA LED scheme, launched in 2015, has distributed over 368.7 million LED bulbs by January 2025 and led to the sale of 407.92 crore (4 079.2 million) LED bulbs in the Indian market, saving approximately 48 terawatt-hours (TWh) of electricity and reducing the retail price of LED bulbs from INR 300-350 (USD 3.80-4.50) to less than INR 80 (USD 1.0) per unit (Press Information Bureau, 2024^[9]). Another example is India's Perform, Achieve, Trade (PAT) Scheme, a market-based white certificate scheme launched in 2012 to reduce specific energy consumption of energy-intensive industries such as iron and steel, cement or textiles. The market-based mechanism enhances its cost effectiveness by certifying excess energy savings, which can be traded by the respective designated consumers. By 2025, the PAT scheme covered over 1 300 designated consumers in energy intensive industries, or about 55% of total industrial energy consumption, and achieved almost 26 Mtoe energy savings (Bureau of Energy Efficiency, 2026^[10]). Furthermore, 9 industry sectors, including steel and refineries, are gradually transitioning into the new Carbon Credit Trading Scheme (CCTS) by 2026-27, while the remaining 4 will continue under the PAT scheme (Ministry of Power, 2024^[11]). In parallel, India's renewable energy capacity continues to expand, with 4.4 gigawatt (GW) capacity allocated to various RE developers in January 2025. This brings the total installed renewable power capacity to around 218 GW by February 2025 (JMK Research & Analytics, n.d.^[12]). However, achieving the government's ambitious target of 50% installed power capacity from non-fossil fuels and reducing emissions intensity by 45% by 2030 will require a significant scale-up in energy efficiency investments.

Market opportunity - scale and limitations

Scale of opportunities

India's energy efficiency market holds immense potential, driven by rapid industrialisation, urbanisation and increasing energy demand. According to estimates by the Bureau of Energy Efficiency (BEE), energy efficiency investments across key sectors could reach INR 1 002 329 to 1 320 630 crores

(USD 121-160 billion) by 2031 (OECD, 2024^[13]). These investments can significantly reduce energy consumption, lower operational costs and support India's climate goals, including achieving net-zero emissions by 2070.

The energy efficiency market in India holds immense scalability potential, yet according to the BEE, only around 130 ESCOs are currently empanelled which have a critical role in implementing energy efficiency measures at their clients and mobilising financing. The energy efficiency market in India is estimated to be worth INR 150 000 crore, whereas Energy Service Companies (ESCOs) only tapped about 5% of this market potential to date (Bureau of Energy Efficiency, 2026^[3]). The following list summarises key areas of opportunities:

- Micro, Small and Medium Enterprises (MSMEs):** As of April 2026, India has over 79 million MSMEs across different economic sectors that are registered with the Ministry of Micro, Small and Medium Enterprises. With around 16.6 million MSMEs (Ministry of Micro, Small and Medium Enterprises, 2026^[14]), the manufacturing sector has been the source of income for 110 million employees, of which 50 million are based in rural areas (Shivprasad, 2026^[15]). As per the Ministry of MSMEs, the sector consumes 25% of industrial energy consumption. The energy consumption of the sector is expected to increase by 50% by 2030, driven by India's economic growth, the emergence and growth of MSMEs, the impact of e-commerce and globalisation, and the ongoing transition to a low-carbon energy system. The sector is critical to align India's growth with a net-zero emissions and inclusive trajectory. Key sectors including foundry, textiles, food processing and brickmaking, currently dominate industrial energy use. More than 90% of enterprises in these sectors are micro enterprises. The definition of MSMEs by size is provided in Table 2.1 below. Targeted interventions and the adoption of energy-efficient technologies can significantly reduce energy consumption while enhancing cost-effectiveness. Upgrading to energy efficient motors of IE3 and IE4 classification, for instance, can achieve 10-15% of savings, installing variable speed drives (VSDs) for motors, and particularly for fans and pumps, can yield 10-15% lower energy consumption during fluctuating load requirements, while the adoption of automated temperature-controlled furnaces can achieve 20-30% savings. While SMEs are key beneficiaries for the implementation of an ESI scheme, its implementation also entails transaction costs for the insurance and validation processes. International experience shows that ESI works best for energy efficiency projects between USD 20 000 and USD 1 million, whereas transaction costs might prevent its cost-effective application for micro-enterprises with investments in energy efficient systems up to USD 20 000. For larger-scale investments above USD 1 million, energy performance contracts based on shared savings can offer a suitable alternative as well.

Table 2.1. Definition of MSMEs by size

Definition of MSMEs by size		
Micro Enterprise	Small Enterprise	Medium Enterprises
Investment in Plant and Machinery or Equipment: Maximum INR 2.5 Crores (USD 0.3 million) and Annual Turnover of up to INR 10 Crores (USD 1.2 million).	Investment in Plant and Machinery or Equipment: Maximum INR 25 Crores (USD 2.9 million) and Annual Turnover of up to INR 100 Crores (USD 12 million).	Investment in Plant and Machinery or Equipment: Maximum INR 125 Crores (USD 14.9 million) and Annual Turnover of up to INR 500 Crores (USD 60 million).

- Industry:** India's industrial sector is a key driver of economic growth, contributing nearly 30% to the nation's GDP in 2020, as reported by the World Bank. With ongoing urbanisation and industrial expansion, energy consumption is set to rise considerably. The International Energy Agency (IEA) estimates that India's energy demand will grow by more than 3% annually

through 2040 (IEA, 2021^[16]) fuelled by increasing manufacturing activity, infrastructure projects and a growing population. This highlights the urgent need for improved energy efficiency strategies to balance economic development with sustainable resource use. Industries with high energy consumption, such as steel, cement and chemicals, present substantial opportunities for efficiency enhancements. The high number of MSMEs in less energy-intensive industrial sub-sectors presents another considerable opportunity for energy efficiency improvements, as outlined above.

- **Buildings:** The building sector is responsible for approximately 35% of India's total energy consumption (Government of India, 2008^[4]), making its contribution to energy efficiency and conservation crucial. Implementing energy efficiency measures such as smart and efficient building design, green construction standards and energy efficient appliances offers significant potential for reducing energy demand. Major factors driving building energy consumption include urbanisation, population growth and the increasing need for space cooling, which is expected to rise sharply as living standards improve and more households gain access to cooling solutions. As per IEA estimates, energy consumption for space cooling in ASEAN countries could almost quadruple by 2040 (IEA, 2022^[17]), emphasising the need for sustainable cooling technologies and energy-efficient building designs and practices.

Challenges

Despite its potential, India's energy efficiency market faces several challenges that complicate the acceleration of energy efficiency improvements and scaling up of investments, particularly with a view to MSMEs. Some of the most important market barriers include:

- Limited understanding of end-users and financial institutions of energy efficiency and other energy-related aspects.
- Competing investment priorities of MSMEs that are more directly linked to their core business and often higher upfront investment costs of EE solutions compared to cheaper, inefficient alternatives.
- Lack of standardised monitoring and verification processes for energy savings complicates their validation and can lead to disagreements.
- Lack of trust of MSMEs regarding the technology provider's ability to achieve the promised energy savings.
- Inconsistent Reporting: Energy usage data across all sectors is often fragmented, making it difficult to benchmark performance.
- Lack of Granular Data: Detailed, sector-specific data on energy efficiency adoption is limited, making market assessments less precise.
- The lack of a standardised mechanism for collecting, analysing and disseminating energy consumption data across various sectors makes it difficult to assess progress and identify investment priorities.
- Limited Transparency in Financing: Information on private sector investments in energy efficiency projects, technology performance and default rates remains scarce, restricting financial institutions from accurately assessing risk and potential returns.
- Limited Access to Finance: There is still limited access to timely and adequate formal credit for MSMEs and ESCOs.
- Shortage of Skilled Labour, including limited availability of certified installers and auditors, as well as accredited ESCOs, which limits the uptake of energy efficiency improvement measures.

- Market Access Constraints for smaller energy technology providers and ESCOs, due to intense competition from large companies and in export markets.

The limited awareness among MSMEs, financiers and insurers about the benefits, potentials and business model of energy efficiency is often combined with a lack of capacity to assess energy efficiency opportunities and their risks, to design and implement energy efficiency measures and monitor the savings achieved, a relatively high perception of risks, as well as limited access to financing and competing investment priorities, particularly for MSMEs.

A strong and supportive policy environment including regulatory measures, information and capacity building as well as incentives for energy efficiency improvements is critically important for the sustainable development of an energy efficiency market in India. The success of financial risk mitigation instruments such as ESI would further require the availability of robust data collection frameworks, real-time monitoring of energy savings and improved market intelligence, combined with a strong collaboration between public and private stakeholders.

3 The potential of MSMEs to become first adopters of an energy savings insurance in India

Micro, small and medium enterprises (MSMEs) form the backbone of India's economy, driving industrialisation, employment generation and economic development. MSMEs promote inclusive growth by creating jobs in urban and rural areas, which helps to reduce regional disparities and limit rural-to-urban migration by strengthening local livelihoods. They contribute to fostering innovation in various industries, and account for about 30% of India's GDP.

The OECD conducted a needs assessment study in 2024, which identified MSMEs as high-impact sector for the implementation of an ESI scheme in India. As of April 2026, India has over 79 million MSMEs across different economic sectors that are registered with the Ministry of Micro, Small and Medium Enterprises, creating jobs for over 350 million people. About 16.6 million or 21% are active in the manufacturing sector, while services account for 36% and trading for roughly 43% (Ministry of Micro, Small and Medium Enterprises, 2026^[14]). As of recent reports, the MSME contribution to GDP stands at around 30% of India's total GDP and 45% of manufacturing output (Bureau of Energy Efficiency, 2024^[1]). Their ability to adapt and innovate has made them a driving force for job creation, skill development and regional economic balance.

As explained in Chapter 3, there are nevertheless obstacles to the contribution of MSMEs to energy efficiency improvement and to the GDP such as information asymmetries, limited access to credit and regulatory challenges. Realising India's vision of becoming a USD 5 trillion economy depends on strengthening this sector.

The BEE strives to align the ESI Scheme with the **“Assistance for Deployment of Energy Efficient Technologies in Industrial Establishments (ADEETIE) Scheme”** of the Ministry of Power, which was launched in 2025. ADEETIE provides support to MSMEs through fiscal financial incentives in the form of interest rate subsidies on loans to implement energy efficient technologies. The ADEETIE Scheme aims to support manufacturing enterprises operating in industrial clusters across 14 energy-intensive sectors: Brass, Bricks, Ceramic, Chemical, Fishery, Food Processing, Forging, Foundry, Glass & Refractory, Leather, Paper, Pharma, Steel Re-rolling and Textile. Based on their energy savings potential and stakeholder interest expressed during the roadmap consultations, the implementation of **pilot projects in selected industrial clusters in the forging, steel re-rolling, food processing and paper sectors appear promising**. At a later stage, further areas such as public or commercial buildings could be explored for the application of the ESI scheme.

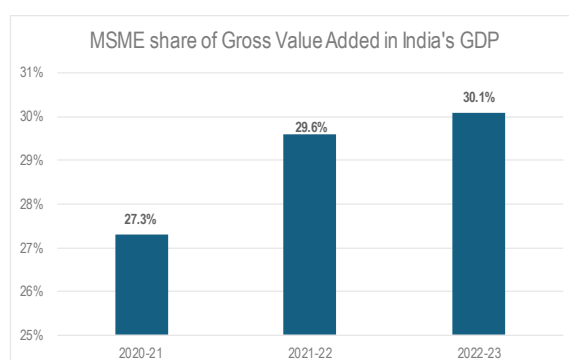
MSMEs operate in two major industrial sectors, namely manufacturing and service industries, engaging in activities ranging from textile production, food processing and engineering goods to IT services, retail and logistics. The Government's procurement policy aims to promote MSME growth by reserving a certain percentage of contracts for MSMEs (Ministry of Micro, Small and Medium Enterprises, 2012^[18]).

Additionally, financial institutions offer specialised loan products and credit guarantees to help these enterprises scale up their operations. Some successful examples include:

- **MSE-GIFT (Green Investment and Financing for Transformation):** As part of the Raising and Accelerating MSME Performance (RAMP) programme, this scheme offered a 2% interest subsidy (rebate) on loans up to INR 2 crores (USD 250 000) for MSMEs adopting green or energy efficient technologies;
- **MSE-SPICE (Scheme for Promotion and Investment in the Circular Economy):** Provides a 25% capital subsidy to MSMEs for adopting circular economy practices.
- **SIDBI financing scheme for energy savings projects:** Targets MSMEs for energy efficiency projects with a minimum loan size of INR 10 lakh (USD 10 000), offering lower interest rates (0.75% below market lending rates) and repayment periods of up to 7 years.

Nonetheless, the market penetration of dedicated financing products remains too limited to have a transformational effect towards higher energy efficiency. Other public programmes including the ADEETIE scheme encourage MSMEs to adopt energy efficient technologies, improve productivity and explore global markets through trade facilitation programs.

Figure 3.1. MSME's share of gross value added in India's GDP, 2020-2022



Source: (Ministry of Micro, Small and Medium Enterprises, 2024^[19])

Over the past five years, MSMEs exports have witnessed a remarkable rise, increasing from INR 3.95 lakh crore (USD 41.5 billion) in 2020-21 to INR 12.39 lakh crore (USD 130 billion) in 2024-25, underscoring their critical role in boosting India's economy and strengthening global trade. The total number of exporting MSMEs in 2024-25 has also increased considerably from 52 849 in 2020-21 to 173 350 in 2024-25. As shown in table 3.1 below, MSMEs demonstrated an exemplary growth trajectory, and contributed a total of 46% to exports as of May 2024, highlighting their growing impact on India's trade performance (Press Information Bureau, 2024^[20]).

Table 3.1. Year wise MSME scale-up under the Udyam Scheme

Year	Micro to Medium	Y-o-Y increase	Small to Medium	Y-o-Y increase
2020-21 to 2021-22	714		3 701	
2021-22 to 2022-23	1 221	+71%	6 476	+75%
2022-23 to 2023-24	1 835	+50%	15 918	+146%
2023-24 to 2024-25	2 372	+29%	17 745	+11%

Source: MSME Udyam Registration Portal and Union budget 2024-25 and 2025-26

4 ESI implementation plan in India

Preparatory phase

Stakeholder engagement and consultations

As part of the preparatory efforts to introduce an ESI scheme in India, the **support from key stakeholders needs to be mobilised to ensure a sustainable anchoring in the Indian market and sufficient uptake of the insurance product**. Key domestic stakeholders for the ESI scheme include the General Insurance Council (GIC), participating banks under the ADEETIE scheme and other interested financial market institutions, insurance companies (to be identified with support from GIC), the Insurance Regulatory and Development Authority (IRDAI), ESCOs and energy efficiency technology providers. The implementation of an ESI scheme further benefits from technical and financial contributions of development partners, multilateral development banks (MDBs), regional development banks and other development finance institutions (DFIs).

Preparatory work to mobilise stakeholder support has commenced through a series of stakeholder consultations conducted between 2024 and 2025. The General Insurance Council (GIC) was constituted in 2001 by the IRDAI under section 64-C of the Insurance Act, 1938. The GIC is an important link between the IRDAI and the non-life insurance industry, and it advocates the industry's interests with the government, and is mandated to facilitate overall growth of the insurance industry in a fair and equitable manner in the interest of all stakeholders. The Council also works with the International Insurance Associations and agencies for information exchange and for the promotion of business collaborations. During the stakeholder workshop in March 2025, the **GIC expressed their support in working with the OECD, the BEE and the ADB to support the design and implementation of the ESI scheme**. The GIC's role as association of insurance companies in India would be to encourage its members to explore the energy efficiency market as a potential business opportunity, and to develop and file an ESI product for registration.

During the roundtable meeting organised on 20th March 2025 in Mumbai, the **IRDAI confirmed its support in approving an ESI product once it is filed by the insurance companies in India and compliant with all applicable regulations**.

Most of the leading banks in India, especially the Small Industries Development Bank of India (SIDBI), ICICI Bank, HSBC Bank, YES Bank, HDFC Bank and the State Bank of India have experience in financing energy savings projects and understand the energy efficiency market. Several banks including HDFC and ICICI already have insurance vehicles (HDFC-ERGO, ICICI Lombard, etc.) and have commitments for financing MSMEs under the ADEETIE Scheme. The BEE could work with these banks to encourage them to (i) provide the guarantee for the performance and the payment, or (ii) work through their insurance verticals to develop and file an ESI product, for instance a surety bond, for approval from IRDAI. If a guarantee is preferred, insurance providers can offer a financial guarantee through the ESCO for fulfilling its obligation as per the agreed terms. If an ESI product in the form of a surety bond is preferred, the GIC could support in expediting the process.

As large domestic banks already have experience with energy efficiency projects, the development of dedicated ESI products through their insurance verticals appears as a promising approach. Surety bonds have been used in many instances for the implementation of ESI schemes, for instance in Colombia and in Europe. The HDFC-ERGO Insurance Company, a vertical of HDFC Bank, is already in discussion with Munich Re for a possible launch of an energy efficiency insurance product. To date, the insurance sector in India has, however, not insured any energy efficiency project.

Needs assessment and priority sectors

A detailed needs assessment aims to gather information on the existing policy and regulatory framework to support the financing and implementation of energy efficiency projects, to assess the readiness of the insurance sector and identify a suitable host agency for the implementation of ESI scheme. The needs assessment also helps in identifying high impact sectors, assessing the energy efficiency financing landscape and identifying potential clients that would benefit most from the implementation of the ESI scheme.

The OECD CEFIM programme, under the guidance of the BEE, conducted a needs assessment study for an ESI scheme in 2024, which identified the MSME sector, particularly the manufacturing industry, as a priority sector for the initial rollout of the scheme. While different industrial sub-sectors dominate in each of India's states and Union Territories, foundry and forging, textiles, food processing and brickmaking are most frequently represented in the roughly 180 MSME clusters across India. Consultation meetings and workshops were conducted between 2022 and 2025 to engage with key stakeholders including financial institutions, insurance companies, ESCOs, MSMEs, technology providers, vendors and international agencies working in India.

During the needs assessment and industry consultation phase, insurance service providers have shown interest in insuring energy saving projects in India. During the consultation phase, Indian stakeholders suggested introducing a hybrid financial de-risking instrument comprising of (i) a credit guarantee provided by an international financial institution supporting the energy efficiency project financing and (ii) an ESI product backing up the agreed energy savings.

Both the ESI scheme and the ADEETIE scheme focus on accelerating energy efficiency improvements in MSMEs in the industry sector. Linking both schemes could enhance their impact, where the participating financial institutions under the ADEETIE scheme finance energy efficiency technologies and measures in SMEs, where an ESI can be included as a performance guarantee mechanism for the project. The details about the ADEETIE scheme are provided in Box 4.1 below.

Box 4.1. Assistance for the Deployment of Energy Efficient Technologies in Industrial Establishments (ADEETIE) scheme

The Ministry of Power has launched the ADEETIE scheme to accelerate the uptake of energy efficient technologies and measures by MSMEs through financial instruments and provide technical assistance in carrying out investment grade energy audits, in preparing detailed project reports, in monitoring and verification of the implementation, and in promotion and awareness raising activities. The scheme provides a 5% (to small and micro enterprises) and 3% (to medium enterprises) interest subsidy on loans from participating financial institutions for technology adoption.

The scheme is designed to help manufacturing businesses in India become more energy efficient. Enterprises operating in MSME clusters across the following energy-intensive sectors are eligible for support: 1. Brass, 2. Bricks, 3. Ceramic, 4. Chemical, 5. Fishery, 6. Food Processing, 7. Forging, 8. Foundry, 9. Glass & Refractory, 10. Leather, 11. Paper, 12. Pharma, 13. Steel Re-rolling, and 14. Textile.

The scheme will be implemented in a phased manner, initially focusing on 60 clusters. In the first three-year phase (FY 2024-25 to FY 2026-27), up to 9 000 MSME units may be covered, followed by another 100 clusters in the second phase. This phased approach allows for demonstrating success and streamlining the scheme based on learnings from the initial phase.

Source: (Bureau of Energy Efficiency, 2026^[21])

A market feasibility study on “**Financial De-Risking Instruments for Energy Efficiency in India**”, conducted by the Asian Development Bank (ADB) in April 2025, recommended the design and implementation of a pilot project for an ESI scheme in the MSME sector. The BEE and the OECD CEFIM programme would work with the ADB and other key stakeholders to support the design and implementation of a pilot.

Identification of pilot clusters

For a pilot implementation of an ESI scheme, MSME clusters have to be determined. The needs assessment study revealed that while different industrial sub-sectors dominate in each of India’s states and Union Territories, foundry and forging, textiles, food processing and brickmaking are most frequently represented in the roughly 180 MSME clusters across India. In consultations with the Bureau of Energy Efficiency, the foundry and paper manufacturing clusters near Coimbatore district in Tamil Nadu and the forging cluster near New Delhi emerged as potential pilot clusters. Both clusters are part of the ADEETIE scheme, have significant energy savings potential and indicated their readiness to participate in the design and implementation of ESI pilot projects. Pilot implementation of the ESI scheme will likely require the mobilisation of additional capital, for instance through multilateral development banks and development partners. The specific design of the pilot projects would be determined by the BEE as facilitator, with technical support from the OECD CEFIM programme and selected ESCOs, and with financial support from domestic and potentially international financial institutions.

Design of the ESI structure

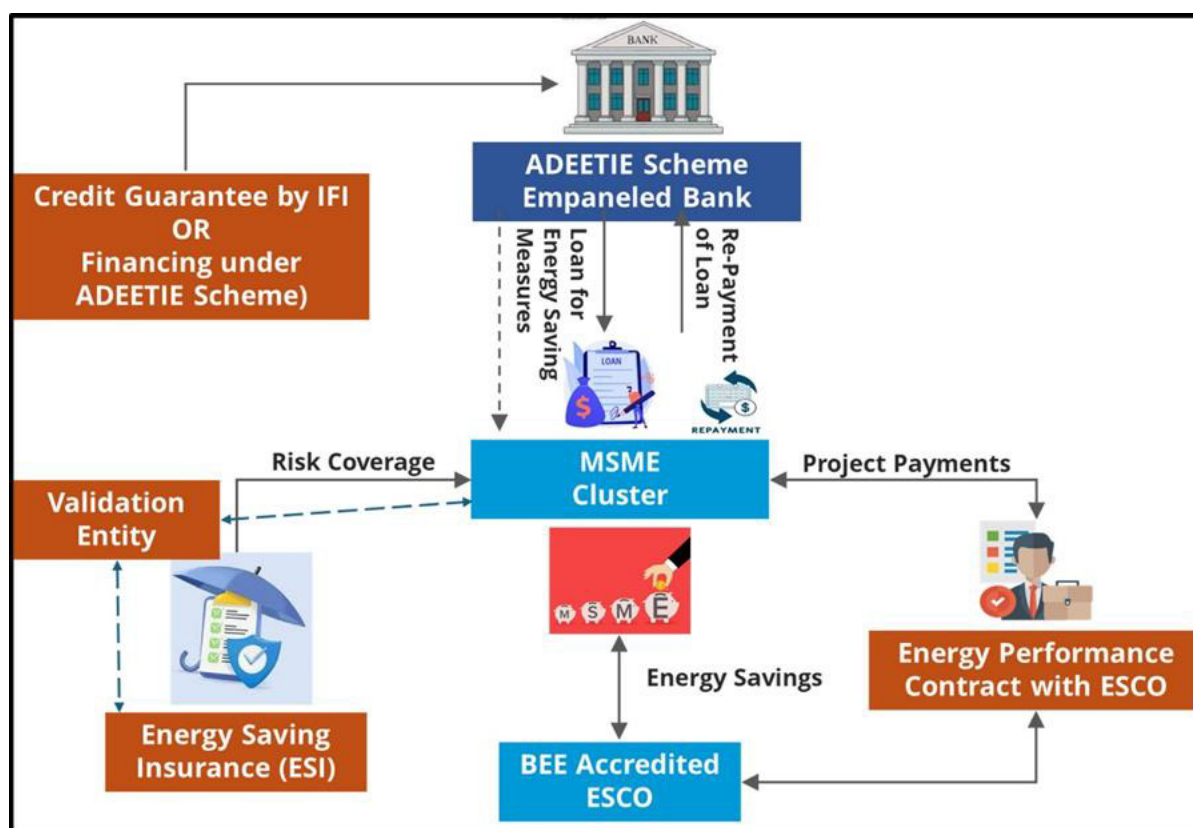
As a next step, the ESI implementation structure needs to be defined and key stakeholders and their responsibilities identified. The proposed ESI implementation structure for India covers financial and non-financial components to mitigate project risks and create investor (bank) and client (ESCOs on behalf

of MSMEs) confidence. The structure explains the roles of the ESI Service Providers, financial institutions, ESCOs, independent validation entity, MSMEs and the mechanism for fund flow.

The ESCO aggregates and designs the energy efficiency project at the cluster level for the participating MSMEs. This aggregation shall ensure the needed scale for financing and insuring energy savings by the banks and insurance companies respectively but also helps to ensure profitability for ESCOs.

A standardised contract signed between the MSME cluster and the ESCO will guarantee the project's performance in terms of energy savings or power generation and is backed by an energy savings insurance product that is activated in case of underperformance. The MSMEs are typically the beneficiaries of the energy savings insurance. Project validation is carried out by an independent validating entity with experience in certification of energy projects, which also acts as an arbitrator with binding decisions for all parties to the contract. Financing may be provided by participating financial institutions interested in creating an energy efficiency lending portfolio. This may include participating banks under the ADEETIE scheme. Empanelled banks under the ADEETIE scheme or more broadly participating financial institutions could receive additional support through (i) government financing under the ADEETIE scheme, or (ii) credit guarantees provided by an international financial institution/MDB. Figure 4.1 presents the proposed ESI implementation structure for India.

Figure 4.1. Proposed ESI implementation structure for India



Note: BEE - Bureau of Energy Efficiency; ESCO - Energy Service Company; ADEETIE - Assistance for the Deployment of Energy Efficient Technologies in Industrial Establishments.

Under the proposed ESI structure, the participating banks can activate the ESI by establishing insurance coverage as a credit requirement. Thereby, the bank enters into a relationship with the MSME/ESCO that

is independent of the client-provider relationship. The energy savings insurance, however, links the two contracts. The ESI reduces the risk of non-payment of the credit, since it guarantees the promised savings. In case the guaranteed savings are not achieved and the ESCO is not able to cover the shortfall, the ESI financially compensates the entity which invested in the energy efficiency project for the shortfall in insured savings.

ESI stakeholders and their role

The effective design and implementation of the ESI scheme in India requires close co-ordination and collaboration between a range of stakeholders, capitalising on their unique mandates and strengths. These include public and private sector agencies, financial institutions, validation entities, ESCOs, individual experts and energy auditors, IT and communications experts, regulatory bodies and international institutions.

The roles and responsibilities of key stakeholders are defined in Table 4.1 below:

Table 4.1. Stakeholder roles and responsibilities for ESI implementation

Stakeholder	Roles and responsibilities
Bureau of Energy Efficiency (BEE)	As statutory body under the Ministry of Power, the BEE is responsible for the overall management of ESI Scheme and the implementation of the ADEETIE scheme. Supported by national and international facilitators, the BEE will provide guidance to all stakeholders to ensure their active participation.
Insurance Regulatory and Development Authority (IRDAI)	A statutory body to assess and approve new insurance products like the ESI product and to protect the interests of insurance policy holders. IRDAI is responsible to set, promote, monitor and enforce high standards of integrity, financial soundness, fair dealing and competence of those it regulates and to ensure the speedy settlement of genuine claims, to prevent insurance fraud and other malpractices and put in place effective grievance redressal machinery.
General Insurance Council (GIC)	The GIC's would work with the BEE and encourage its member insurance companies to develop and file an ESI product for approval with IRDAI. GIC would assist in expediting the process for product design, registration and approval. The GIC was constituted in 2001 by IRDAI under section 64-C of the Insurance Act, 1938. GIC is an important link between IRDAI and the non-life insurance industry and represents the insurance industry in dealing with the Government.
Insurance companies	Interested non-life insurance service providers as members of GIC would collaborate with the BEE, GIC, and other stakeholders in India to customise and file an ESI product for approval with IRDAI. With support from the BEE and facilitators like the OECD, participating insurance companies should organise capacity building for their staff members regarding energy efficiency projects and validation of energy efficient technologies and programmes.
Re-insurance service providers	Re-insurance companies can play an important role in building the confidence of insurance companies while entering a new market or expanding their business. Re-insurance providers like MunichRe can assist local insurance companies in introduction of ESI product.
Commercial banks and other financial institutions	The financial sector is expected to play a key role in financing energy efficiency projects of MSMEs. More than 26 banks and financial institutions have registered under the ADEETIE Scheme of the BEE to provide priority financing to participating MSMEs and are expected to strengthen their co-ordination and exchanges, including to reduce the need for collateral. The ESI as the performance guarantee for the energy saving projects can replace the requirements of collaterals.
Technology providers	Energy efficient technology providers supply and install appliances and equipment to achieve improved energy performance at their clients, i.e. MSMEs. The BEE identified more than 190 technologies, which can be procured by MSMEs for improving energy efficiency. The technology providers shall ensure the agreed level of performance of the supplied technologies and provide services for repair and maintenance if required.
Energy Service Companies (ESCOs)	ESCOs play multiple roles in ensuring the successful implementation of ESI scheme. ESCOs sign energy performance contracts with MSMEs (at cluster level), secure a bank loan, conduct an energy audit to establish the baseline and implement the agreed measures. They regularly report on achieved energy savings, and co-ordinate with technology providers and the independent validation entity. ESCOs would also be responsible in strengthening their knowledge and capacity in handling ESI-related aspects.

SMEs	SMEs would be the clients of energy performance contracts, that want to invest in energy-efficient technologies to increase their productivity while reducing their energy bills and GHG emissions. Some might be interested in becoming frontrunners in implementing high-performance technologies. They are also beneficiaries of the insurance coverage.
Validation entity	An independent and credible validation entity needs to be determined to act as both an evaluator and validator of the energy efficiency project design and implementation. It further acts as an arbitrator in case of disagreement between clients and technology providers/ESCOs, and validates the savings achieved if an insurance claim is filed.
Facilitator	An institution with extensive knowledge about the ESI scheme and its implementation will take the role of a facilitator. The facilitator provides guidance to the BEE and other stakeholders during all stages of ESI scheme implementation and supports the co-ordination of all stakeholders for the successful rollout of the scheme. This role could for instance be played by OECD, ADB or IIEC.

Implementation Phase

Facilitating entity

Based on a thorough market and needs assessment and further preparatory work, the design of the ESI scheme needs to be aligned with existing policies and regulations, and complementary to existing programmes and initiatives. A significant amount of technical assistance and capacity building for banks and insurance companies, but also ESCOs and technology providers is typically required. **Determining a single facilitating entity is beneficial to co-ordinate involved stakeholders, guide the process and provide technical assistance** for vetting the key parameters at different stages of design and implementation. Either the OECD, with support from key international and local institutions, or a dedicated Indian institution could facilitate the rollout of the ESI scheme in India.

Proposed timeline - 1 Month after the ESI Roadmap launch

Independent validation entity and definition of monitoring and verification methodologies

Project validation plays a vital role in ensuring credibility and accuracy of energy savings calculations for energy efficiency projects. Under the ESI model, an independent third-party review validates that measurement reports comply with established guidelines and reflect actual performance. This process enhances transparency and trust by objectively assessing the reported savings based on an agreed methodology and confirming that calculations are based on a well-defined baseline and relevant operational factors.

The BEE could select a recognised and trusted validation entity, assigning it a mandate to establish a credible energy consumption baseline before project implementation and to conduct the validation in the case of a claim filed by the ESCO or MSME for not achieving the agreed energy savings. In case energy efficiency is a new area of work for such validation entity, (i) training and capacity building for their validation team members might be required, and/or (ii) the validation entity might opt to hire BEE-certified energy auditors and managers as validators. **The validation entity would be in charge of defining the monitoring and verification methodologies for the technical validation of energy savings, as well as all the mechanisms, templates and forms that guide the implementation of the scheme.** Common validation methodologies are derived from internationally recognised procedures, such as the IPMVP or ISO.

Proposed timeline - Months 1-3

Standardised energy performance contract

The BEE could work with ESCOs, financial institutions, GIC and the development partners, regional and multilateral development banks and other development finance institutions and international organisations providing technical assistance (e.g. the OECD and ADB) to develop a standardised and simplified energy performance contract. This standardised energy performance contract should be developed and vetted by a law firm experienced in the field of energy, mortgage and insurance to ensure compliance with domestic regulations and legal practices. The contract should reflect the roles and responsibilities of the parties to the agreement, the terms for the technology selection and supply, installation and maintenance. The standardised contract should further include an energy savings guarantee and clearly define the roles of the insurance product and the independent validation entity.

The contract should establish the responsibilities of the technology provider regarding the supply and installation of equipment and clearly define the agreed performance guarantee in terms of promised savings or energy generation. On the client's (MSMEs) side, the contract determines the terms for timely payment, the provision of access to the facilities and appropriate equipment maintenance. Among the most significant features of the contract are:

- Contract duration
- Savings guarantee
- Reference energy tariffs
- Periodicity of measurements
- The monitoring, verification and validation methodology
- The insurance activation criteria
- The indemnification mechanisms for unrealised savings, and
- Dispute resolution

Proposed timeline - Months 2-4

Energy savings insurance product

The successful implementation of the ESI model will rely on the interest and commitment of insurance companies and financial institutions to adopt an energy savings insurance product and finance energy efficiency products on condition of the adoption of such an insurance. The insurance is meant to strengthen the confidence of ESCOs, technology providers and MSMEs in the achievement of the contractually agreed energy savings commitments. The ESI contract between the ESCO and the insurance company aims to mitigate financing risks and strengthen the confidence of commercial banks and other financial institutions in providing the loans. Here, the ESI could play an important role in eliminating the requirements of many banks for collateral when providing the bank loans. The BEE could work with the GIC and its members to ensure that the ESI product is designed and filed for approval with IRDAI. Based on experience from ESI implementation in other markets, surety bonds or bank guarantees are suitable options for an energy savings insurance product, while most ESI products rely on surety bonds. Furthermore, requirements and procedures for linking the ESI with financial incentives provided under the ADEETIE scheme needs to be defined.

Proposed timeline - Months 2-4

Capacity building and awareness raising campaign

The successful adoption of the ESI scheme will require training and capacity building of the key stakeholders and experts involved in its rollout in India. The BEE could work with the OECD, ADB and other international organisations, bilateral development partners, regional and multilateral development banks and other DFIs to design effective training programmes and awareness raising campaigns. These should aim to strengthen the skills of financial institutions and insurance providers in underwriting energy efficiency projects, and of ESCOs in developing and implementing effective energy efficiency measures and technologies, but also to inform the key stakeholders about the functioning of the scheme and to create momentum for its uptake. Multiple channels should be leveraged to engage key stakeholders including ESCOs, energy auditors, technology providers, commercial banks and other financial institutions, the independent validation entity, insurance and reinsurance service providers, MSMEs, Public Sector Undertakings (PSU) and the state agencies. A series of workshops, roundtables, webinars, TV and radio advertisements and other digital tools could be used to raise awareness about energy efficiency and the ESI model, while dedicated training programmes could serve to strengthen capacities.

Proposed timeline - Constant process starting from Roadmap launch and continuing for 12 months

Pilot project design

Correctly designing and implementing at least two pilot projects are key for rolling out the ESI scheme in India. This component is the combined outcome of all other components and should be designed in parallel to other ESI components. Based on the BEE's prioritisation of sectors and industrial clusters, two pilot projects in two industrial clusters in India will be developed in detail. During the pilot phase, the BEE, with support from the facilitator and financier would guide the key stakeholders on the different stages of implementing the ESI scheme. This will ensure a seamless and straightforward application of the scheme in the pilot projects. A multilateral or regional development bank like the ADB could be brought in to fund the preparatory work and pilot implementation of the ESI scheme in the selected clusters. This would include the drafting and adaptation of standardised energy performance contracts and other documents, establishing the baseline energy consumption and equipment usage, co-ordinating with banks, insurance companies, ESCOs, technology providers and other stakeholders, conducting the verification and validation of the first pilot projects. After the signing of energy performance and ESI contracts, and after obtaining a loan where needed, pilot implementation would start. While pilot projects aim to attract a high level of visibility at national and international level, the outcomes and lessons learned would help in integrating the ESI scheme with the pipeline of projects under the ADEETIE scheme.

Proposed timeline - Months 2-6

Regular monitoring and verification

Besides ensuring appropriate operation and maintenance procedures for the installed equipment, a regular monitoring and verification of the achieved savings helps to mitigate risks of defaults and shortcomings in savings. Any shortfall in savings during the pilot testing and during the initial years of implementation may lead to the loss of confidence of key stakeholders. If a savings shortfall during the initial years is correctly identified by the MSMEs/ESCOs and confirmed by the validation entity, the provider should either take technical correction measures or compensate the client (MSME). Should the provider fail to compensate the client, the insurance would be triggered to financially compensate the beneficiary according to the contract terms. Thereby, the insurance backing should ensure continued trust in the model despite a

confirmed shortfall in achieved savings. The definition of appropriate and detailed monitoring, reporting and verification standards and processes is therefore of utmost importance.

The validation entity, in close co-ordination with the technology provider and the insurance company, should monitor the actual accrued energy savings on an annual basis and utilising the agreed methodology. This should avoid any requirements for a premature economic compensation due to underperformance of the technologies.

Technology providers should rely on the monitoring methodology and process agreed in the standard energy performance contract. Smart data loggers or meters can serve to collect information about the real-life use of the equipment and achieved savings.

In order to validate the achieved savings compared to the committed savings and for maintaining proper records, the following documents should be submitted by the technology provider and the ESCO to the validation entity and the insurance provider:

- Data and photo of the data-logger (including date and time of the data collection) against the parameters as defined in the energy performance contract for each equipment type for each year of the contract. This process should be repeated on the same date every year to ensure consistency.
- Calibration certificate of the equipment and the data-loggers or independent verification report to ensure their accuracy.

This process should be part of the validation methodology and should be performed in close consultation with the validation entity. The standard contract should clearly and in detail define the methodologies for measurement and verification. An online data management system or platform could help in recording and streamlining relevant data.

Annual monitoring and verification of energy savings using the agreed methodology should be ensured, as well as a final project evaluation and validation at the end of the energy performance contract. This will help in determining the overall savings achieved, as well as the payback period of the investment and recording of any profits after the completion of the ESI and ESCO performance contracts.

Proposed timeline - validation of project design before start of implementation, validation of installation and annual M&V of savings; final project evaluation and validation

Learning framework

Regularly document the implemented energy efficiency measures and achieved savings, as well as the lessons learned from project implementation across different sub-sectors. Based on this data, develop case studies to disseminate learnings with national and international stakeholders and mobilise the further uptake of ESI products in India.

Proposed timeline - Starting from implementation and at regular intervals

ESI rollout

Based on the learnings from the pilot projects, the BEE with technical support from the facilitator, could develop a long-term strategy to incorporate the ESI scheme fully into phase II of the ADEETIE scheme, to support financial institutions and insurance companies in developing a project pipeline for energy efficiency projects and rolling out the ESI scheme across India. The ESI model also has significant potential for implementation in the public sector, particularly as a key element in green procurement processes. Public

sector actors in India are often familiar with using instruments like surety bonds and performance contract surety bonds to cover financing risks, although these have not been linked to energy savings performance in the past. The standardised contracts and independent validation create transparency and predictability of energy savings expectations and build accountability into public procurement processes. Therefore, integrating the ESI into public sector financing frameworks could be considered at a later stage to support enhancing energy efficiency in the public sector. The potential to roll out the ESI scheme to further sectors could be explored as well.

Implementation steps and timelines

Major implementation steps for the ESI scheme along with the implementation timelines are provided in Table 4.2 below. These steps summarise the entire process covering the design, implementation and monitoring of the ESI scheme in India. Additional steps can be added after deriving learnings from the pilot implementation in India.

Table 4.2. Implementation steps and timelines, preparatory and implementation phase

Steps	Activities	Timeline
Preparatory Phase		
Step 1	Consultation meetings with key stakeholders to understand their readiness, capacity and knowledge about the ESI model, and ensure their participation	January 2023 to June 2025
Step 2	ESI Needs Assessment, including identification of * priority sectors * priority clusters for pilot implementation	May-October 2024
Step 3	Design ESI structure in consultation with key market stakeholders	March-June 2025
Implementation Phase		
Step 1	Determine facilitating entity	M1 after launch
Step 2	Determine independent validation entity and methodologies for verification and validation	M 1-3
Step 3	Prepare standardised energy performance contract and templates for recording energy savings, M&V and validation of projects; s	M 2-4
Step 4	* Design the ESI product and submit it to IRDAI for approval * Determine the procedure for alignment with the ADEETIE scheme	M2-4
Step 5	Develop capacity building and awareness raising campaign	
Step 6	* Develop detailed pilot project design (e.g. consult shortlisted energy efficiency projects under the ADEETIE scheme) * Sign energy performance contract and ESI contract, submit loan application for pilot projects * start implementation	M2-6
Step 7	Conduct regular monitoring and verification, as well as validation of technology performance and achieved savings	After project implementation and annually
Step 8	Learning framework: * Regularly document implemented measures and achieved savings, as well as lessons learned * Develop case study examples to disseminate learnings with national and international audience	Starting from implementation and at regular intervals (monthly/every 6 months)
Step 9	Roll out the ESI scheme across India and consider its expansion to further sectors, e.g. public buildings	TBC

Note: M – month; TBC – to be confirmed;

5 Facilitating the ESI implementation

Establishment of the energy consumption baseline

Reliably determining the baseline energy consumption is the first step to pursue energy efficiency or sustainability goals in a targeted manner. A good baseline doesn't just help with tracking energy savings; if the data is trustworthy and accessible, it can further help in identifying areas for improvement and in securing bank loans. Four fundamental steps are required to establish the energy consumption baseline:

- **Define the Metrics** - The ESCO and validation entity are required to define metrics to collect the energy baseline data effectively. Metrics will help in comparing the baseline with future energy consumption over a period of time.
- **Normalise the Data** - The next step is to normalise the baseline data to create a trustworthy comparison between the current and future years.
- **Set up Real-time Monitoring** - It is the real-time data that allows facilities to be proactive in driving down energy intensity and costs.
- **Sub-meter Specific Processes** - For medium to large MSMEs, it is important to capture the data at sub-meter level to ensure accuracy.

Energy management baseline metrics

Table 5.1 provides the metrics for recording the baseline parameters to help establish a strong and reliable baseline of energy use.

Table 5.1. Energy management baseline metrics

Parameters	Description
Electricity Consumption	Electric energy used in each period, measured in kilowatt-hours (kWh) - Utility bills, available historic data or typically measured at the main meter.
Energy Consumption	Total energy used from all sources in a given period (e.g. electric + gas + Renewables), reported in a common metric (e.g. kWh, BTUs or Therms).
Energy Intensity	Energy consumption divided by a normalization factor (e.g. kWh/unit or BTUs/ft ²).
Energy Costs	Total dollars spent on energy in a given period (USD). This accounts for the timing of energy usage and other factors, in addition to the amount of energy used.
Energy Costs v/s Budget	Variance between energy expenses and budgeted amount in a given period (USD).
Energy Cost Intensity	Total dollars spent on energy in a given period per unit during the entire period of service provided (USD/unit).
Sources of Energy Wastage	This includes checking insulation, inspecting HVAC systems, assessing lighting fixtures, windows, doors, boilers, and other systems installed in a building or industry.

Scheduling and Human Resources Log	Prepare a log providing the scheduling details of the use of various equipment and appliances, working hours of building/industry, list of workers/employees including their duties and duty hours.
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Normalisation of energy use data

A normalisation process is required for putting the data into context and enabling valid comparisons. For example - in MSME operations, it is critically important to understand energy consumption in relation to production. Measuring production-normalised energy intensity allows businesses to understand the true meaning of their energy consumption. A proper project scoping exercise should attempt to determine which factors have the biggest impact on energy consumption. In many industrial environments, production is the most important normalisation factor, while weather conditions, for instance expressed in cooling degree days, have an impact on the electricity use for space cooling. Table 5.2 below provides the required normalisation approaches. ESCOs should use them for accurate calculation of energy savings.

Table 5.2. Baseline data normalisation approaches

Approach	Description
Multi-year averaging	Averaging multiple years' data to create a more reliable baseline of energy consumption.
Production normalisation	Dividing energy usage by a unit of throughput to determine the energy intensity of production.
Weather normalisation	Dividing energy usage by a unit that quantifies the effects of weather, e.g. Cooling Degree Days.
Space normalisation	Dividing energy usage by the area of a facility (e.g. ft ²).
Volume normalisation	Dividing energy usage by the volume of a facility (e.g. ft ³). Commonly used in temperature-controlled facilities.
Revenue normalisation	Dividing energy usage by the revenue of a product line, facility, or business to determine the efficiency with which energy is used to create value. Commonly used to compare facilities with different production approaches.

Real time monitoring

The most frequently used method for setting up the baseline is to simply tally up the utility bills to get a snapshot of monthly or annual energy spend. This approach is simple and inexpensive, but the data collected through this method isn't actionable. It does establish a baseline, but it doesn't indicate:

- When energy usage is highest
- When rates and demand charges are highest
- Which systems or equipment have the highest energy consumption

In most states of India, the commercial and industrial customers don't get charged a flat rate for electricity. Most are on some form of time-of-use rate, whereby rates vary by time of the day, week and/or year. Depending on the contract, these rates may be known well in advance. It can be very helpful during the energy baseline process to understand when rates are spiking, or penalties are being applied. It may make sense to schedule maintenance during these times or even shut down the operations entirely.

Through setting up a real-time data collection system, facilities get a much clearer picture of their own patterns of energy use during working hours versus off-shift usage. This helps flag areas for improvement.

Large spikes in consumption can point the way to targeted improvements, whereas high usage across the board may point to a more capital-intensive strategy.

The utility meters with Advanced Metering Infrastructure (AMI) data are not always a good substitute for real-time monitoring systems. AMI systems or “smart meters” are typically installed by utilities to make meter reading easier, but it is still insufficient for effective energy management, primarily because of the time lag in receiving data. Thus, real time monitoring data helps in developing a strong baseline.

Sub-metering of specific areas for more accurate data

Typically, power usage data is collected at the main meter, which does not provide enough detail for industrial operations and does not take into account potential billing errors. An energy auditor would want to double check and uncover power factors and overvoltage issues that can cost money and damage equipment. It is useful to install sub-meters to monitor the most energy-intensive processes within a facility.

Baselining specific systems in addition to whole facilities allows for a much more targeted energy management approach. While deciding on the installation of sub-meters, the boundaries should be drawn around processes rather than equipment. Sub-metering should account for all major components involved in the process and should feed into one or more of the metrics defined for the production-normalised energy intensity.

Economic analysis of suitable energy efficient technologies

Various energy-efficient technologies can be effectively financed under energy efficiency programs. In the MSME sector, which spans diverse industrial and commercial subsectors, certain technologies stand out for their high energy-saving potential. These commonly include efficient air conditioning systems, high-efficiency motors, modern boilers, advanced cooling solutions, and both thermal and photovoltaic solar systems. When replacing conventional low-efficiency systems, these technologies can lead to substantial reductions in energy consumption and operational costs. Table 5.3 below outlines selected technologies and their estimated energy savings potential within the Indian context. The table is an amalgamation of data and information gathered from various reports published by the Bureau of Energy Efficiency and the performance reports published by various ESCOs in India. The list is not exhaustive, and additional relevant technologies and measures may also be considered based on specific applications and sectoral needs. The implementation of the ESI model in the pilot projects and beyond may be based on a smaller number of selected technologies with high savings potential.

Table 5.3. Suitable energy efficient technologies and their savings potential

Suitable technologies and measures	Savings potential (%)
Screw Compressor-based Chiller With VFD	40%
Oil sealed screw vacuum pump with VFD	35%
Reducing AHU pressure drops	33%
Installation of Microturbines	30%
Variable Volume Ratio (VVR) technology	23%
Installation/ Overhauling of the recuperator to reduce the flue gas heat loss	20-25%
Waste Heat Recovery (WHR) system in Stenters	20-25%
Energy efficient thermic fluid heater	20%
Replacing the present pumps with proper head pumps	20%
Scroll Compressor-based Chiller	20%

Energy efficient boiler system	15%
Online Chiller Condenser Automatic Tube Cleaning System	12%
Installation of energy-efficient roller motor for each rolling strand	10-15%
Installation of IE3 Motors	10-15%
Overhauling of the furnace with proper insulation lining	10-15%
Installation of Anti-friction Roller Bearing in Rolling Mill Strands	10%
Mist Cooling System for Cooling Towers	10%
Installation of Energy Management System (EMS) to optimize the power consumption	5-15%
Fluidized Bed Combustion Boiler	4%
Insulation of Boiler Surface	3-4%
Installation of Oxygen and CO sensor to optimize the excess air of the furnace	2-4%
Microprocessor Based Occupancy/Load Scheduling	5-10%
PM Screw compressor with waste heat recovery	5%

Note: VFD – variable frequency drive; VVR – variable volume ratio; WHR – waste heat recovery; EMS – energy management system

Source: (Bureau of Energy Efficiency, 2026^[22])

Implementing these energy efficiency measures and technologies involves an associated investment, which can vary significantly based on factors such as the capacity, expected output and the specific nature of the intervention. Costs are also influenced by the choice of brand, type of technology, equipment size, efficiency rating and the supplier or service provider. Table 5.4 below presents indicative and generalised cost estimates for a selection of these measures; however, actual costs may differ depending on the unique requirements and conditions of each implementation.

Table 5.4. Average reference cost for energy efficiency technologies for MSMEs across sectors

Technology option	Indicative investment cost: small-scale projects (INR)	Indicative investment cost: large-scale projects (INR)
Screw Compressor-based Chiller With VFD	INR 10 to 15 lakh per unit	INR 25 to 40 lakh per unit
Oil sealed screw vacuum pump with VFD	INR 10 to 15 lakh per unit	INR 25 to 40 lakh per unit
Reducing AHU pressure drops	INR 1 to 3 lakh per modification	INR 5 to 8 lakh per modification
Installation of Microturbines	INR 50 to 75 lakh per unit	INR 1 to 1.5 crore per unit
Variable Volume Ratio (VVR) technology	INR 5 to 10 lakh per system	INR 15 to 25 lakh per system
Installation/ Overhauling of the recuperator to reduce the flue gas heat loss	INR 10 to 15 lakh per unit	INR 25 to 40 lakh per unit
Waste Heat Recovery (WHR) system in Stenters	INR 20 to 30 lakh per system	INR 50 to 80 lakh per system
Energy efficient thermic fluid heater	INR 2 to 5 lakh per heater	INR 10 to 15 lakh per heater
Replacing the present pumps with proper head pumps	INR 1 to 3 lakh per pump	INR 5 to 10 lakh per pump
Scroll Compressor-based Chiller	INR 5 to 8 lakh per unit	INR 15 to 20 lakh per unit
Energy efficient boiler system	10-15 lakh per boiler	25-40 lakh per boiler
Online Chiller Condenser Automatic Tube Cleaning System	INR 2 to 4 lakh per system	INR 8 to 12 lakh per system
Installation of energy-efficient roller motor for each rolling strand	INR 3 to 6 lakh per motor	INR 10 to 15 lakh per motor
Installation of IE3 Motors	INR 50,000 to INR 1 lakh per motor	INR 2 to 5 lakh per motor
Overhauling of the furnace with proper insulation lining	INR 5 to 10 lakh per furnace	INR 15 to 25 lakh per furnace
Installation of Anti-friction Roller Bearing in Rolling Mill Strands	INR 1 to 3 lakh per set	INR 5 to 8 lakh per set
Mist Cooling System for Cooling Towers	INR 3 to 5 lakh per system	INR 10 to 12 lakh per system
Installation of Energy Management System (EMS) to optimize the power consumption	INR 50,000 to INR 1.5 lakh per EMS	INR 2 to 5 lakh per EMS
Fluidized Bed Combustion Boiler	INR 15 to 20 lakh per boiler	INR 40 to 60 lakh per boiler
Insulation of Boiler Surface	INR 1 to 2 lakh per boiler	INR 4 to 6 lakh per boiler

Installation of Oxygen and CO sensor to optimize the excess air of the furnace	INR 2 to 5 lakh per set	INR 8 to 12 lakh per set
Microprocessor Based Occupancy/Load Scheduling	INR 1 to 2 lakh per system	INR 3 to 5 lakh per system
PM Screw compressor with waste heat recovery	INR 15 to 25 lakh per unit	INR 40 to 60 lakh per unit

Note: VFD – variable frequency drive; VVR – variable volume ration; WHR – waste heat recovery; EMS – energy management system; large-scale projects meaning covering investment costs between USD 30 000 and USD 150 000.

Source: (Bureau of Energy Efficiency, 2026^[21])

The economic viability of technologies largely depends on their frequency of use, which influences how quickly equipment costs are recovered. Equipment used continuously tends to yield faster payback compared to systems used only during work hours or daylight. Intensive use is more common in large-scale operations with multiple work shifts. Maintenance costs, which vary by equipment type and age, also play a significant role. Additionally, electricity and fuel prices impact cost-effectiveness.

Validation process and requirements

Project validation plays a vital role in ensuring credibility and accuracy of energy savings calculations for projects using energy-efficient equipment. It involves an independent third-party review to verify that measurement reports comply with established guidelines and reflect actual performance. This process enhances transparency and trust by objectively assessing the reported savings and confirming that calculations are based on a well-defined baseline and relevant operational factors. By aligning with the principles of the ESI scheme, validation ensures consistency, reliability and compliance throughout the project lifecycle. Under the standardised contract, a two-stage validation process should be required, including: A. project validation and B. Arbitration process.

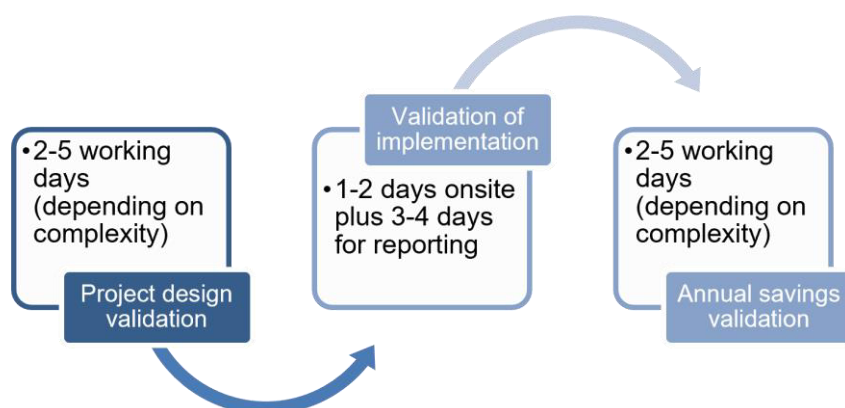
A. Project validation

- Project validation at contractual stage.
- Installation validation, to be conducted when the project equipment is installed and at the end of commissioning stage to check that everything is working correctly and according to the contract.
- Annual validation during the insured period: at the end of each year, the provider reports on the savings achieved. The validation entity could be asked to review these reports. In cases of underperformance or disagreement between the parties, a review must be requested; in such cases, however, the arbitration process would already have been initiated.

B. Arbitration process

- It can be started in cases of underperformance or disagreement between the parties on the annual savings report issued by the technology provider.
- The validation process begins when a technology provider identifies a project opportunity or a client is seeking to upgrade to more energy-efficient equipment. As outlined above, the validation is required at three key stages during the contracting period (Figure 5.1):

Figure 5.1. Project validation stages and estimated timeframes



The estimated time required by the validator to complete each validation task is calculated based on the assumption that all required documentation is submitted correctly and in full for a single-equipment project. The actual duration of the validation process may vary depending on the complexity of the technology or the specific project requirements and can be mutually agreed upon between the client and technology provider.

I. Role of validators

The validator plays a crucial role in the ESI scheme, serving as the key authority for creating and setting benchmarks as well as for auditing and ensuring project compliance with the initial benchmarking standards.

Required qualifications of validators

Validators are required to possess a degree or diploma in engineering, a background in accountancy, or a related field like insurance or surveying. Additionally, passing the relevant examinations conducted by the Insurance Institute of India (III) and completing a practical training period under a senior validator are required. The BEE certified Energy Auditors and Managers can also qualify for the validators' job as they carry the necessary knowledge. Their qualifications include:

- **Sector-specific knowledge**
 - In-depth understanding of the industry in which the technology provider operates.
 - Familiarity with energy efficiency technologies and project benchmarking standards in general as well as for the target sectors.
- **Certification & professional credentials**
 - Must have passed the exam from the Insurance Institute of India (III) or must be a certified energy auditor with the BEE.
 - Proven experience in conducting energy audits and verifying energy-saving calculations
- **Technical & analytical skills**
 - Proficiency in validating project documentation, measurement reports and savings calculations.
 - Ability to analyse data logger records and assess energy performance
- **On-site inspection capability**

- Experience in conducting site visits to verify equipment installation and operation
- Knowledge of data logging systems and ability to reset and manage measurement period
- **Regulatory & compliance knowledge**
 - Understanding of relevant energy efficiency policies, standards and compliance requirements
 - Ability to ensure that projects align with initial benchmarking and contractual obligations

A qualified validator ensures the integrity of the validation process, providing reliable and objective assessments to uphold the credibility of the ESI program.

II. Validation of project design

The project design validation focuses on verifying the estimated energy savings calculated by the technology provider based on the specific project conditions and boundaries.

To determine the estimated energy savings, the energy consumption of the existing equipment is measured and compared to the projected consumption of the new, validated equipment. This comparison accounts for key operational factors such as usage hours and other relevant parameters. As part of this validation step, project operation factors are established to enable an accurate assessment of energy savings after one year. Project operation factors are typically set in the validation methodologies and agreed between the provider and the client. They serve as an input for the validation entity. Once the committed energy savings value is validated, clearance is given for the activation of the ESI contract. This contract will include all relevant operational factors and the verified committed energy savings. The energy savings calculation methodology for the annual verification of energy savings and any potential insurance claim needs to be agreed. To validate the project scope, proposed equipment and calculated energy savings, the technology provider must present the following documents:

- **License documentation**
 - Proof that the technology provider is authorised to sell the specified equipment.
 - Confirmation that the technology provider is licensed to install the equipment or is using a certified contractor for installation.
- **Project documentation**
 - Detailed project documents, along with supporting documents and standards for each equipment. Furthermore, the calibration certificates of all equipment and measuring equipment needs to be disclosed.
 - Verification that the project design scope meets ESI eligibility criteria and aligns with the defined Project Operation Factors.
- **Measurement reports**
 - Reports based on the ESI measurement report template, validating the accuracy of the technology provider's measurement process.
 - Assessment of savings calculations to ensure accuracy and completeness.
- **Committed energy savings calculation**
 - Computation of committed energy savings as defined under the ESI scheme.
 - Verification of calculation accuracy, alignment with measurement reports and consistency with manufacturer specifications.
- **Signed ESI Contract between the ESCO/MSME and the Insurance Service Provider**
 - A legally binding contract that includes all agreed terms and conditions.

These documents serve to uphold transparency, accuracy and compliance throughout the ESI validation process. Once the technology provider submits all required project documents, the validator reviews the information, verifies the findings and validates the committed energy savings in the form of a report, the ESI contract is officially activated.

III. Verification and validation of project implementation

The project implementation or installation validation involves a site visit conducted by the validator to verify that the correct equipment has been installed as per the project documentation and the equipment is fully operational. During this visit, the validator performs a visual inspection to confirm proper installation and functionality of the system. Additionally, the validator examines the mandatory data logger, which is installed by the technology provider to facilitate the recording of key operational parameters as defined in the design stage. Upon completion of the site visit, the validator issues an installation validation report, documenting the implementation status and ensuring compliance with project requirements. This report verifies that the technology and data logger have been correctly installed in accordance with project requirements. The following documents are required for implementation validation:

- **Final project documentation**
 - As outlined in the ESI contract, each equipment type should adhere to the initial design scope.
- **Manufacturer's datasheet of installed data logger**
 - Provides technical specifications to confirm the eligibility of the data logger.
 - Valid calibration certificate or independent verification report
- **Confirmation that the data logger meets ESI methodology requirements and proper calibration.**
 - The data logger calibration must be valid for the upcoming annual measurement period.

These documents collectively ensure that the implementation/installation meets ESI standards, allowing for accurate performance monitoring and savings validation.

IV. Annual validation of the project

After the end of each year, the provider reports on the energy savings achieved to support the annual validation of the savings. This is based on the same operating factors outlined in the ESI contract. This process determines whether economic compensation is necessary due to underperformance of the installed technology. To facilitate this process, the client (MSME/ESCO) and the technology provider must agree on the following factors during the design stage:

- The defined units of transformed energy
- The predetermined cost per energy unit
- The committed energy savings value, calculated as per the agreed ESI methodology

This structured approach ensures that actual savings align with initial projections, providing a clear basis for performance evaluation.

V. Validation for insurance claim

The validation of an insurance claim ensures that it aligns with the baseline established during the initial project assessment. The insured must initiate the claim, after which the insurer appoints a qualified validator to review it. This process may include an on-site inspection to verify the claim's legitimacy. The key steps in the validation process are as follows:

- Review of claim and past reports: the validator examines the claim alongside previous validation reports, including design, implementation and annual assessments.
- On-site inspection and documentation: if required, the validator conducts a walkthrough audit to assess the current condition of the project or equipment.
- Energy savings verification: the reported energy savings are compared against the committed values, referencing implementation and annual validation reports.
- Compliance assessment and decision: any deviations from the contract terms are identified and the claim is either approved or rejected based on the agreement set during implementation.
- Investigation of performance issues: if energy savings do not meet expectations, the validator determines the underlying cause and may initiate a separate inquiry if needed.

The following documents are mandatory for the insurance validation process:

- Project design validation report and associated documents
- Project implementation validation report and associated documents
- Legal and valid contract between the insurer and the insured party
- Annual validation report and associated documents (if applicable)

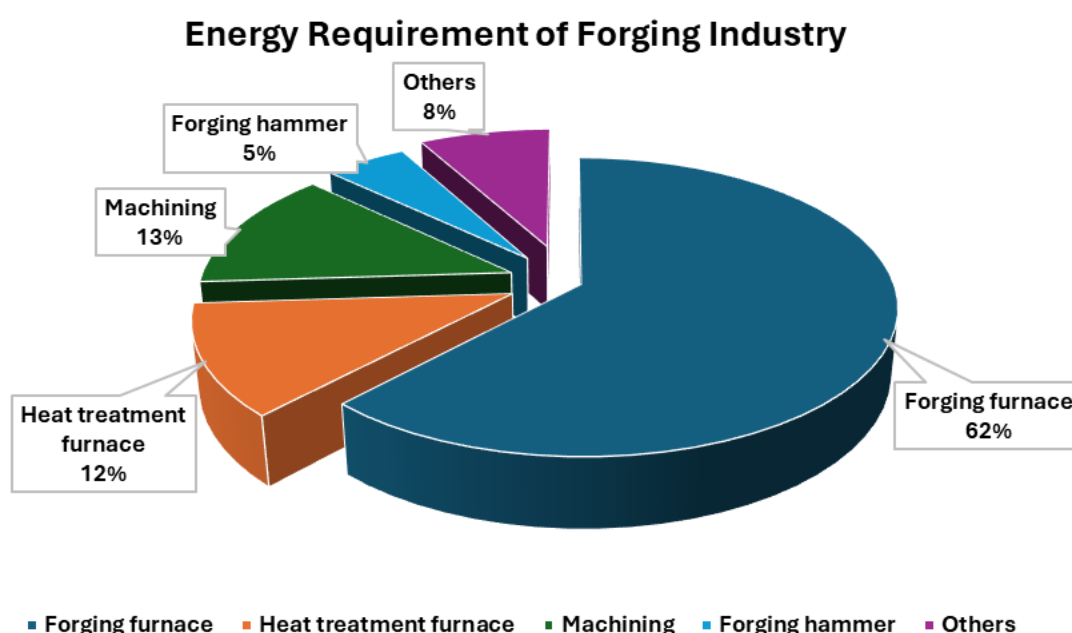
The validator has the sole authority to approve or reject the claim based on a valid assessment report. However, if the insured disputes the decision, they may challenge it on merit, as per the arbitration clause outlined in the contract between the insurer and the insured. Normally, the standardised contract stipulates that the opinion of the validation entity should be accepted by the parties so that the decision of the arbitration is not questioned.

6 Financial modelling and pilot business case

To enhance the understanding of key stakeholders in India, a business case is prepared for a forging MSME cluster near New Delhi. The forging industry in India is a vital pillar of the country's manufacturing ecosystem, significantly supporting key sectors such as automotive, power and general engineering. Forging, which involves shaping metal using compressive forces, produces a wide range of high-strength components like crankshafts, camshafts, gear parts, axle beams and industrial valves.

The forging sector faces persistent challenges such as volatile steel and industrial fuel prices, rising input costs and increasing competition from imported components - particularly from China, Japan, Korea and Europe. Forging operations typically involve energy-intensive processes such as furnace heating, hammering and machining, powered by fossil fuels and electricity. Additionally, many units employ heat treatment furnaces and compressed air systems, further increasing energy demand. With appropriate technology upgrades and efficiency interventions, the sector can achieve significant energy savings, making it more sustainable and cost-effective.

Figure 6.1. Energy consumption in the forging industry (2023)



Source: (Bureau of Energy Efficiency, 2021^[23])

Table 6.1 provides the list of possible options for energy efficiency projects that can be implemented along with the Energy Savings Insurance models in forging clusters of MSMEs.

Table 6.1. Short, medium and long-term energy efficiency measures

Short-term	Medium-term	Long-term
Optimise the overall loading of furnaces by better production planning.	Install kWh indicator cum integrator for induction furnace. The kWh indicator cum integrator combines the functionality of a kWh indicator (which shows the current energy consumption) with an integrator (which keeps a running total of energy used over time)	Install air preheater for the combustion air supply to the heat treatment furnaces & heating furnace
Improve combustion efficiency of heating & heat treatment furnace	Use of translucent sheets to maximise use of day light	Replace existing oil-fired heating & heat treatment furnaces with gas fired furnaces
Operate furnaces at the optimum temperature	Use of Eco Ventilators of hot air exhaust	Provide ceramic fibre insulation for batch-operated furnaces
Reduce heat losses from furnace openings	Use of automation for temperature control in forging and heat treatment furnace	Practice oxygen enrichment in furnace
Maintaining the correct amount of furnace draft	Improve the overall insulation levels and close the openings in furnaces, so as to minimize heat losses	Segregate high pressure and low pressure compressed air users in the forging industry
Arrest compressed air leakages by vigorous maintenance	Install Automatic Power Factor Control (APFC) System	Install variable frequency drive for the screw compressor
Optimize overall operating pressure of compressors based on the system requirement	Replace faulty capacitor banks and relocate capacitors to the machine ends, or from the Main Switch Boards (MSBs) to the Sub-Switch Boards (SSBs) (at the substation ends), to minimise voltage drop in cables	Replace pneumatic operated tools with electrical tools and use energy efficient equipment in allied operations
Provide ball valves at the user ends of compressed air cleaning hoses	Install Automatic - Star - Delta - Star (ASDS) converter in the lightly loaded motors which handle fluctuating loads. ASDS makes use of the fact that whenever the load on the motor is less (<40%), it switches the motor to star mode, and as soon as the load increases (>40%), it switches the motor to delta mode.	Use of online O ₂ measurement and control system
Install blow air nozzle for cleaning applications involving compressed air	Install automatic voltage stabilizers for lighting circuits and other precision electronic circuits	Install energy efficient reheating furnace with flat roof
Replace the delta connection with permanent star in case of motors, which are lightly loaded. Lightly loaded motors (below 50% of full load), it is generally recommended to permanently connect it in a star (or wye) configuration rather than delta. This is because a star connection reduces the voltage applied to each winding, leading to lower current draw and improved energy efficiency.	Install lighting transformers in all major lighting feeders and operate the lighting circuit at 210V	Use of energy efficient motors and air compressors

To explore opportunities for improving energy efficiency in the forging industry and to identify potential sustainable interventions, **three strategic projects have been selected for in-depth evaluation**. These

initiatives were chosen based on their practicality, relevance and the significant energy savings they can bring to typical forging operations.

- The first initiative focuses on replacing traditional furnaces with energy-efficient models, which generally provide superior insulation, enhanced combustion control and improved thermal efficiency. Such upgrades can lead to substantial reductions in fuel consumption and minimised heat losses, resulting in both lower operational costs and reduced emissions. This process of replacing the furnace can save up to 25% of energy (as per the published data of EE furnaces).
- The second project involves switching from conventional fossil fuel-powered furnaces to induction furnaces. Induction furnaces use electricity and offer precise temperature control, faster heating cycles and a cleaner process without direct combustion emissions. These furnaces provide an environmentally friendly alternative that aligns well with modern forging operations and can save up to 15-20% energy in the process.
- The third project aims at upgrading to IE3 (Premium Efficiency) motors, replacing standard motors. These high-efficiency motors operate at a significantly better energy performance level compared to older or less efficient counterparts, reducing electrical energy use, enhancing system reliability, and lowering operating temperatures. This improvement not only cuts energy costs but also extends equipment life and reduces the frequency of maintenance and can provide energy savings of up to 20%.

Together, these three projects promise considerable energy and cost savings while also presenting scalable, replicable solutions that can be adopted by small and medium-sized forging enterprises. The specific savings potential in each case may slightly vary, depending on the chosen technology, operating conditions and possible interactions with other energy efficiency measures that are being implemented. Implementing these initiatives would lead to enhanced productivity, a reduced environmental footprint and contribute positively to India's broader goals of energy security and sustainable development.

Financial assessment

The financial assessment of energy efficiency (EE) projects in the forging sector follows a structured methodology, beginning with clearly defined baseline assumptions. These are based on the typical operations of a single unit operating daily mirroring the working conditions commonly found in the Faridabad forging cluster near the Delhi NCR region. All the below calculations are done for a single unit. Establishing these parameters is essential for accurately estimating both the financial benefits and the environmental impact of the proposed energy-efficient upgrades. The cost savings calculations are presented in Table 6.2 below.

Table 6.2. Financial Model with proposed energy savings in the forging industry

Parameters	Unit	Value	Source
Industry Performance Parameters			
Single furnace capacity	Kg	200	Technology Compendium by UNIDO
Daily working hours	Hr	10	
Yearly working hours	Day	300	
Furnace oil energy content	kWh/litre	10.5	Assumption
Energy savings measure	°	°	
Average potential energy savings	%	21	Calculated based on average potential energy saving of all three proposed energy saving measurements. The background data is taken from DPRs published in SAMEEESHASHA Portal by Bureau of Energy

			Efficiency
Expected investment	INR	47 000 000	Calculated
Financial parameters	°	°	
Electricity tariff	INR/kWh	12	Average landed price including all charges and taxes
Grid emission factor for furnace oil	tCO ₂ /MWh	.0716	
Grid emission factor for electricity	tCO ₂ /litre	0.003	
Debt equity	%	70:30:00	Assumption
Interest rate with insurance	%	10	
Interest rate without insurance	%	12	

Source: (Bureau of Energy Efficiency, 2021^[23])

Table 6.3. Energy efficiency measures, expected savings and payback periods

Energy efficiency measures	Estimated energy savings(kWh)	Expected investment (INR)	Expected savings (INR)	Payback for individual measures
Replacement of furnace with energy efficient option	262 500	2 500 000	1 080 000	2.31
Replacement of furnace with induction furnace	175 000	2 000 000	720 000	2.8
Replacement of motors with IE3 motors	19 564	200 000	234 768	0.9

Note: The electricity tariff is considered at INR12 per kilowatt-hour (kWh), inclusive of fixed charges, energy charges and applicable taxes. This rate serves as the basis for calculating reductions in electricity expenses resulting from the adoption of energy-saving technologies.

- On the environmental front, emission reductions are calculated using a grid emission factor of 0.716 tonnes of CO₂ per megawatt-hour (tCO₂/MWh) for electricity, and 0.00296 tCO₂ per litre for fossil fuels. These values help quantify the decrease in greenhouse gas emissions attributable to lower energy consumption.
- From a financing perspective, the project assumes a capital structure composed of 70% debt and 30% equity. This is a common approach that allows for leveraging external funding while still requiring investment from the project owner.
- The debt component carries an interest rate of 10% under an insurance-covered scenario and 12% without insurance. The repayment of this debt is spread over a period of 10 years, while the insurance is considered to be active for the first 5 years of the project's lifecycle. This is aligned with the experience on ESI coverage in other countries. These financing terms directly influence annual cash flows, particularly in terms of interest payments and insurance premiums.
- On the depreciation front, two approaches are factored into the analysis. The straight-line depreciation method (SLM) is applied at a rate of 10% per year, which spreads the cost of the equipment evenly over its useful life. For taxation purposes, a higher depreciation rate of 15% is used, which helps reduce taxable income in the earlier years of the project.
- A salvage value of 10% is assumed at the end of the equipment's life, which reflects the expected residual value of the assets after full depreciation.
- The corporate tax rate is set at 30%, which is standard and influences the net savings after tax deductions.

In addition to these costs, annual operation and maintenance (O&M) expenses are estimated at 2% of the initial investment cost, which is a recurring cost that slightly reduces the annual financial savings. In India,

O&M costs for energy efficiency projects can vary based on the technology, size and location of the project, but typically range from 1% to 2% of the initial capital expenditure.

Committed energy and financial savings

With all these parameters in place, the next step is to estimate the yearly cost savings generated from reduced energy consumption. These savings are calculated by comparing the baseline energy use with the expected consumption after project implementation. The resulting annual gross savings are then adjusted for O&M costs, loan repayments and taxes to derive the net cash inflow for each year. These cash flows represent the project's financial benefit over its lifetime.

To assess the long-term viability and attractiveness of the investment, financial metrics such as Net Present Value (NPV), Internal Rate of Return (IRR), and the payback period are computed.

Table 6.4. Payback period and IRR of the project at unit level

Key financial indicators	With Energy Savings Insurance	Without Energy Savings Insurance
NPV (INR)	3 650 197	3 604 412
IRR	12.26%	12.05%
DSCR	2	2
Payback (Years)	5.4	5.7

Note: NPV – Net present value; IRR – Internal rate of return; DSCR – Debt service cover ratio

- NPV is calculated by discounting future cash flows to their present value using an appropriate discount rate, typically aligned with the cost of capital or the required rate of return. In this case, the NPV is INR 36 50 196, indicating that the project is expected to generate a positive return above the cost of capital.
- The IRR, which stands at 12.26%, reflects the effective annual return expected from the project and is a critical metric in investment decisions - especially when comparing different project alternatives. Since the IRR exceeds common lending rates and inflation levels, it suggests that the project offers a financially sound investment opportunity.
- The payback period, which is around 5.4 years, indicates the time it will take for the project to recover the initial investment from the net cash flows, which is considered reasonable given the 10-year project horizon.
- The average Debt Service Coverage Ratio (DSCR) is also 2 which signifies a strong financial position.

When the analysis is scaled to the cluster level - representing approximately 400 forging units with a combined annual production capacity of around 2 million metric tonnes, which closely reflects the estimated scale of forging industry clusters - the project continues to demonstrate strong financial viability. By maintaining the same baseline assumptions used at the unit level, the aggregated evaluation at the cluster level confirms the economic feasibility of implementing the energy efficiency measures across the industry.

Table 6.5. Payback period and IRR of the project at cluster level

Key Financial Indicators	With Energy Savings Insurance	Without Energy Savings Insurance
NPV (INR)	33 310 871 966	33 158 255 704
IRR	28.92%	28.65%

DSCR	3	3
Payback (Years)	3.12	

Note: NPV - Net present value; IRR - Internal rate of return; DSCR - Debt service cover ration

- The Net Present Value (NPV) of the project, calculated under these parameters, remains positive, indicating that the initiative is expected to deliver returns exceeding the cost of capital over its lifespan. This reinforces the financial attractiveness of the intervention.
- Furthermore, the Internal Rate of Return (IRR) is estimated at 28.92%, a robust figure that highlights the project's capacity to yield substantial annual returns. This rate significantly surpasses typical borrowing costs and prevailing inflation rates, suggesting that the project not only adds value but also offers a competitive edge over other investment options.
- The payback period is calculated at approximately 3.12 years, which means the initial capital invested in the project would be recovered through savings within a relatively short timeframe and this time frame is shorter from the previous due to the scale of the project - and is well within the 10-year life span of the project. This short payback horizon enhances the project's appeal, particularly for small and medium enterprises seeking quicker returns on their investments.
- Additionally, the average Debt Service Coverage Ratio (DSCR) stands at 3, indicating a strong ability to meet debt obligations comfortably. A DSCR of this magnitude reflects healthy cash flow generation relative to debt service requirements, further strengthening the case for investment from financial institutions and other stakeholders.

Overall, the financial indicators at the cluster level affirm that this energy efficiency initiative is not only environmentally beneficial but also economically sound, making it a strategic opportunity for scaling sustainable industrial practices across the forging sector in India.

In conclusion, this financial assessment provides a comprehensive overview of the economic performance of the proposed energy efficiency project in the forging sector. It not only quantifies the direct financial benefits but also highlights the potential for carbon emission reductions, making a strong case for both economic and environmental sustainability. The use of detailed assumptions and a step-by-step methodology ensures that the analysis is robust, transparent and well-suited for decision-makers evaluating similar energy interventions. By covering the risk that the technology provides or if the ESCO fails to achieve the committed energy savings, the ESI can facilitate access to financing for energy efficiency improvements and may also reduce the cost of financing, for example through concessional credit lines and reduced interest rates for insured projects.

7

Next steps

The following steps will be taken to support the rollout of the ESI scheme in India:

- The BEE will launch the ESI Implementation Roadmap in alignment with the ADEETIE scheme in 2026 to operationalise the ESI scheme.
- The BEE will convene a Technical Committee (TC) comprised by main actors for the implementation of the ESI scheme to coordinate implementation activities and ensure alignment with the ADEETIE scheme. The TC should then convene regular meetings under the chairmanship of the DG-BEE.
- The design and implementation of ESI pilot projects, and the creation of a pipeline of projects could be coordinated by the TC with guidance from the BEE.
- The BEE, supported by the facilitating entity and other partners, should (i) identify capacity building needs of key stakeholders and define targeted training programmes, and (ii) develop standardised templates, contracts and other documents.
- Based on the learnings from an initial rollout of both the ESI scheme and the ADEETIE scheme, the BEE could develop a long-term strategy for embedding the ESI scheme in phase II of ADEETIE.

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Annex A. ESI and EEI worldwide

This annex lists all/selected energy savings insurance (ESI) and energy efficiency insurance (EEI) projects across all the continents. Please see Table A A.1 below.

Table A A.1. Energy Savings Insurance and Energy Efficiency Insurance (EEI) products

S.N.	Country	Continent	Name of insurance company involved	Name of financial institution involved	Sectors covered
ENERGY SAVINGS INSURANCE (ESI)					
1	Morocco	Africa	Local insurance companies such as Saham Assurance, Wafa Assurance, AtlantaSanad, RMA Watanya, and AXA Assurance Maroc	Société d'Ingénierie Énergétique - SIE	Small and Medium-sized Enterprises (SMEs)- 98% of all enterprises in Morocco
2	Mongolia	Asia	Tenger Insurance	XacBank	MSMEs in all Economic sectors
3	India		Munich RE, HDFC ERGO and General Insurance Council are in the advance stages of designing energy saving insurance product	The Bureau of Energy Efficiency (BEE), with support from the OECD's CEFIM Programme and ADB, is actively working on the development of ESI via ongoing workshops and stakeholder consultations with ESCOs, auditors, and technology providers.	Micro, Small and Medium-sized Enterprises (MSMEs)
4	Indonesia		PT Zurich Asuransi Indonesia, PT Maskapai Reasuransi Indonesia (Marein) designing reinsurance product, General Insurance Association of Indonesia (AAUI) and Indonesia Sharia Insurance Association (AASI) is designing standards.	PT Sarana Multi Infrastruktur (PT SMI), PT Bank UOB Indonesia, Indonesia Eximbank	Industrial and Commercial and Service Sector
5	Croatia	Europe	GoSafe is promoting the ESI initiative, without the direct involvement of insurance companies. However, prominent insurers such as Croatia osiguranje d.d. and Grazer Wechelseitige Versicherung (GRAWE) are notable in the landscape.	Croatian Bank for Reconstruction and Development (HBOR), European Bank for Reconstruction and Development (EBRD), other commercial Banks	Small and medium sized enterprises
6	Greece		Local insurance companies provide performance-based insurance products	Centre for Renewable Energy Sources and Saving - CRES	Industrial, commercial and residential
7	Italy		Atradius, Reale Mutua	Banco BPM, Civibank	Hotels, Food processing and Food retail
8	Portugal		ABARCA SEGUROS, AGEAS Portugal	CréditoAgrícola/Santander/Novo Banco/Millennium BCP	Hotels, Textile
9	Slovakia		Local insurance companies provide performance-based	Slovak Innovation and Energy Agency - SIEA	Hotels, Food and Food Processing

			insurance products		Industry
10	Spain		Crédito y Caución, CESCE, Credito y Caución	ABANCA/Santander/(PF4EE)/iberCaja/Triodos Bank	Hotels and Food processing
11	El Salvador	North America	Aseguradora Suiza Salvadoreña - ASESUISA	Banco de Desarrollo de la República de El Salvador - BANDESAL/Banco Azul/ ANCOVI/Banco Hipotecario/Banco Industrial	Small and Medium-sized Enterprises (SMEs)
12	Mexico		Local insurance companies offer performance-based insurance products that guarantee projected energy savings	Fideicomisos Instituidos en Relación con la Agricultura - FIRA, Inter-American Development Bank (IDB)	Agricultural and Agro-Industrial
13	Argentina		Local insurance companies offer performance-based insurance products that guarantee projected energy savings	Bank for Investment and Foreign Trade - BICE	Small and Medium-sized Enterprises (SMEs)
14	Brazil		Kovr Seguradora	State of Rio Grande do Sul - BRDE/State of Espírito Santo - Bandes/State of Goiás - e Goiás Fomento	Industrial, Agriculture and Energy-intensive SMEs
15	Chile	South America	Grupo de Inversiones Suramericana - SURA/CESCE	BancoEstado	Hospitality, Agroindustry, Vitiviniculture, and Fisheries
16	Colombia		Grupo de Inversiones Suramericana - SURA	Bancóldex/Banco Davivienda	SMEs in Hospitality, Healthcare, Manufacturing, and Commercial, Services
17	Paraguay		Local insurance companies offer performance-based insurance products that guarantee projected energy savings	Agencia Financiera de Desarrollo - AFD	Industrial sector
18	Peru		Local insurance companies offer performance-based insurance products that guarantee projected energy savings	Corporación Financiera de Desarrollo - COFIDE	Hotels, Hospitals & Clinics, Food Processing, Fisheries, and Textile
ENERGY EFFICIENCY INSURANCE					
19	United States	North America	The Hartford Steam Boiler Inspection and Insurance Company (HSB) - Munich-re/Xicato/XL Group	Regional Development Banks	Commercial Buildings, Industrial Energy Efficiency, HVAC Systems, Energy-Saving Retrofits
20	United Kingdom	Europe	HSB- Munich Re/Zurich Insurance	Local Financial Institutions	Energy Efficient Technologies, Commercial Buildings, Industrial Energy Efficiency
21	Ireland		HSB- Munich Re/Zurich Insurance	Local Financial Institutions	
22	France		Agence AXA International	BNP Paribas	Green Building Projects, Energy Efficiency in Industry

Annex B. International case studies

This annex provides a review of selected international case studies on ESI and EEI. They include:

- Name, type of project and location of case study
- ESI or EEI
- Name of the partner bank and the implementing agency/partners
- Achievements
- Impact
- Related project references

A. Energy savings insurance

Eight case studies on the implementation of the energy savings insurance in Colombia, Chile, Morocco and Mongolia are presented below.

Energy savings insurance for SMEs in Colombia

With the support of the Inter-American Development Bank, and in collaboration with BANCOLDEX, Colombia has become the first country in Latin America and the Caribbean to issue energy saving policies under the Energy Savings Insurance Program (ESI) to the benefit of small and medium-sized enterprises that have invested in energy efficiency and power generation projects.

Case study A1: Hospital Méderi

Location: Colombia

Programme: Energy Savings Insurance (ESI) Program

Partners: Inter-American Development Bank (IDB), BANCOLDEX

Background: Hospital Méderi, the largest hospital network in Colombia, recognised the need to address its growing demand for reliable hot water across its Bogotá facility. To modernise and put in place a more sustainable solution, the hospital embarked on a project to install a new thermal system. With the support of the ESI programme, the institution aimed to find a trusted technology provider and achieve the desired outcomes (IDB, 2019^[24]).

Project description:

- **Technology implemented:** the project at Hospital Méderi involved the installation of **220 solar collectors** on the hospital's rooftops to guarantee a continuous supply of hot water and fulfil the 24/7 needs of hot water in the patient rooms and other hospital areas.
- **Total investment: USD 200 000** was allocated for the complete overhaul, installation and integration of the new system.

- **Energy source transition: the replacement led to a drastic reduction in the consumption of natural gas**, transitioning a majority of the hot water heating load to a cleaner and more sustainable source (BASE, 2024^[25]).

Achieved benefits:

- **Annual energy savings:** The hospital achieved guaranteed energy savings of approximately USD 23 000, leading to a payback period of nine years (BASE, 2024^[25]).
- **Operational efficiency:** The new heating solution provided a reliable and consistent hot water supply, improving the comfort and service quality for students, staff and faculty. It also reduced maintenance demands and enhanced system longevity.

Impact:

- **Support for SMEs and green tech providers:** The project fostered collaboration with local SMEs offering energy-efficient technologies and installation services, further promoting green economic development.
- **Replication potential:** The success of this initiative underlined the replicability of energy efficiency projects backed by insurance mechanisms, making it easier for other institutions to follow suit with confidence in their return on investment (IDB, 2019^[24]).

Case study A2: Villa Martha Hotel

Location: Colombia

Programme: Energy Savings Insurance (ESI) Program

Partners: Inter-American Development Bank (IDB), BANCOLDEX

Background: The **Villa Martha Hotel**, a boutique hospitality business in Colombia, recognised the rising cost of electricity and the environmental impact of its energy consumption. As part of its commitment to sustainability, guest comfort and cost-efficiency, the hotel sought to integrate renewable energy into its operations.

Hospitality businesses, especially smaller, family-run establishments like Villa Martha, often face financial and technical barriers when investing in clean energy. The ESI Program, supported by the IDB and BANCOLDEX, provided the hotel with the tools and security to confidently invest in solar energy by offering technical validation and performance guarantees (IDB, 2019^[24]).

Project description:

- **Technology implemented:** the hotel installed a **10 kilowatt-peak (kWp) photovoltaic (solar PV) system**, designed to offset a significant portion of the hotel's electricity demand.
- **System specifications:**
 - Rooftop-mounted photovoltaic panels
 - On-grid configuration with net metering to feed surplus power back into the local electricity grid (BASE, 2024^[25]).
- **Total investment: USD 30 000**, covering system design, equipment procurement, installation and technical validation (IDB, 2019^[24])

Achieved benefits:

- **Electricity cost reduction:** the PV system significantly reduced the hotel's reliance on grid electricity, cutting energy bills and protecting the business from future tariff increases.

- **Energy independence:** the solar system improved energy resilience, especially important in regions susceptible to grid instability.
- **Insurance-backed performance:** through the ESI Program, the Villa Martha Hotel was able to secure **performance insurance**, ensuring that the expected energy savings from the solar system would be met or losses compensated. This drastically reduced investment risk and was a key enabler for project financing (BASE, 2024^[25]).

Impact:

- **Tourism and sustainability alignment:** the Villa Martha Hotel positioned itself as a **sustainable tourism** destination, appealing to eco-conscious travellers and strengthening its market differentiation.
- **SME empowerment:** the project demonstrates how small businesses can harness **renewable energy** with the right financial and risk-mitigation tools in place.
- **Scalability:** The success of this project shows that even small-scale installations, when supported by the right mechanisms, can make a meaningful impact on both operational costs and environmental performance (BASE, 2024^[25]).

Case study A3: Neiva Plaza Hotel

Location: Neiva, Colombia

Programme: Energy Savings Insurance (ESI) Program

Partners: Inter-American Development Bank (IDB), BANCOLODEX.

Background: the **Neiva Plaza Hotel**, a well-established business in the heart of Neiva, faced rising operational costs due to its aging and inefficient boiler-based water heating system. The existing method was reliant on a 24-year-old pyro-tubular boiler that was fuelled by liquefied petroleum gas (LPG). The system was generating nearly 6 000 hot water litres per day for laundry, kitchen and showers, resulting in high levels of pollution and operating costs (IDB, 2019^[24]). Through Colombia's **Energy Savings Insurance (ESI) Program**, the hotel found the ideal platform to confidently invest in new clean energy technology with guaranteed performance outcomes.

Project description:

- **Technology implemented:** the hotel replaced its old **fossil-fuel boilers** with a **solar thermal water heating system**, a clean and efficient technology suitable for hotels with high hot water demand.
- **System capacity:** the project consisted of replacing the old boiler with **22 solar collectors** installed on an area of 61.7 m² with five backup heat pumps, resulting in the reduction of gas consumption by 70%. With each collector producing between **10 000 and 12 000 British thermal units (BTU) per hour**, the hotel achieved efficient heat production for a 6 000-litre tank, maintaining a temperature range of 50 to 55 degrees Celsius (BASE, 2024^[25]).
- **Total investment:** **USD 25 000** was invested in the project, covering equipment, installation, and engineering services (IDB, 2019^[24]).

Achieved benefits:

- **Energy savings:** The new solar thermal system achieved an impressive **70% reduction in energy consumption** compared to the previous boiler setup. This translated into major utility cost savings.

- **Cost savings:** The investment of **USD 25 000** yielded guaranteed energy savings of approximately **USD 8 000 per year**.
- **Simple payback period:** Thanks to high energy savings and relatively low investment, the project's **simple payback period was less than 3 years**, making it a highly attractive financial decision (BASE, 2024^[25]).

Strategic impact:

- **Sustainability commitment:** the Neiva Plaza Hotel elevated its reputation as an environmentally responsible business in the tourism sector, aligning with growing guest expectations for green hospitality practices.
- **Cost control and profitability:** the reduced energy costs directly contributed to improved profitability and allowed the hotel to reinvest savings into other service areas.

Case study A4: Agrícola El Retorno S.A. - Chile

Location: Chile

Programme: Energy Savings Insurance (ESI) Program

Sector: Agriculture

Partner: Inter-American Development Bank (IDB)

Background: Agrícola El Retorno S.A., a farming enterprise in Chile, was facing increasing energy expenses due to its high reliance on grid electricity for daily operations. In pursuit of long-term sustainability and cost stability, the company decided to invest in clean energy. Through the **Energy Savings Insurance (ESI) Program**, the company accessed performance-guaranteed technology, risk mitigation, and financing mechanisms to install a solar energy system on its premises.

Project description:

- **Technology implemented:** installation of an **on-site solar photovoltaic (PV) system** to generate clean electricity for agricultural operations.
- **Capacity: 63 kW of capacity** designed to meet a significant portion of the farm's daytime energy demand.
- **Total investment: USD 134 328** covering the design, purchase and installation of the solar PV system (Netto et al., 2020^[26]).

Achieved benefits:

- **Annual energy savings:** the system produces approximately **107 600 kWh per year**, reducing dependency on grid electricity.
- **Annual cost savings:** estimated at **USD 11 768 per year**, contributing to long-term operational cost reductions.
- **Simple payback period:** approximately **12 years**, after which the company continues to benefit from nearly free solar power (Netto et al., 2020^[26]).
- **Energy resilience and independence:** the solar system offers greater control over energy costs and supports business continuity during power outages or price spikes.

Impact:

- **Sustainable agriculture leadership:** Agrícola El Retorno's project positions the company as a leader in sustainable farming, demonstrating how clean energy can be integrated into agriculture.
- **Boost for the clean tech market:** the initiative supported local technology providers and installers, enhancing growth in Chile's distributed solar energy sector.
- **Model for replication:** With its straightforward implementation and measurable returns, this solar PV project stands out as a **replicable model** for other SMEs in the agricultural sector across Latin America (Netto et al., 2020^[26]).

Case study A5: Ariztía El Paico poultry processing plant

Location: El Paico, Chile

Programme: Energy Savings Insurance (ESI) Program

Sector: Food processing/poultry industry

Partner: Inter-American Development Bank (IDB)

Background: Ariztía, a leading poultry processing company in Chile, sought to enhance energy efficiency at its **El Paico processing plant**, one of its most energy-intensive facilities. With rising electricity costs and growing environmental commitments, the company turned to the **Energy Savings Insurance (ESI) Program** to support a high-impact industrial upgrade, focused on replacing aging electric motors with newer, high-efficiency models.

Project description:

- **Technology implemented:** replacement of **three large-capacity electric motors** to improve process efficiency and reduce electricity consumption.
- **Motor capacities:** 600 kW, 522 kW and 336 kW
- **Total investment:** **USD 80 279** for procurement, installation, and commissioning of the energy-efficient motors (Netto et al., 2020^[26]).

Achieved Benefits:

- **Annual energy savings:** the new motors deliver **energy savings of 43 319 kWh per year**, reducing the plant's electricity footprint.
- **Annual cost savings:** the plant benefits from approximately **USD 46 223 in cost savings annually**, significantly improving its operating margins.
- **Simple Payback Period:** Less than **2 years**, making it a financially attractive investment with rapid returns (Netto et al., 2020^[26]).
- **Improved Operational Reliability:** The upgraded motors reduce downtime and maintenance costs, enhancing overall production efficiency.

Impact

- **Industrial energy efficiency showcase:** Ariztía's motor replacement project serves as a prime example of **how industrial process upgrades** can yield rapid financial and environmental returns.
- **Risk reduction via ESI:** the ESI mechanism **mitigated performance risk**, giving Ariztía confidence in the expected savings and enabling a smooth financing process.

- **Inspiration for sector-wide adoption:** the project sets a precedent for energy efficiency improvements in **Chile's broader food processing sector**, encouraging similar investments across the industry (Netto et al., 2020^[26]).

Case study A6: Bayas del Sur S.A.

Location: Chile

Programme: Energy Savings Insurance (ESI) Program

Sector: Food processing/agroindustry

Partner: Inter-American Development Bank (IDB)

Background: Bayas del Sur S.A., a prominent company in the Chilean agro-industrial sector known for processing berries and other fruit-based products, was seeking to modernise its production infrastructure. The company identified its **aging steam boiler** as a major source of energy inefficiency and operational cost. To reduce fuel consumption and lower emissions, Bayas del Sur engaged with the **Energy Savings Insurance (ESI) Program** to de-risk investment in a new, energy-efficient boiler system.

Project description:

- **Technology Implemented:** installation of a new energy-efficient industrial boiler to replace an older, less efficient system.
- **Capacity:** 3 tons of steam, sufficient to meet the plant's process heating demands more efficiently.
- **Total Investment:** USD 230 100, covering the full equipment upgrade and installation (Netto et al., 2020^[26]).

Achieved benefits:

- **Annual energy savings:** the new boiler generates approximately **665 000 kWh in energy savings per year**, significantly reducing fuel consumption.
- **Annual cost savings:** the company now saves an estimated **USD 39 000 per year** on energy costs, contributing directly to operational profitability.
- **Simple payback period:** less than **6 years**, after which the investment continues to generate annual savings and operational efficiencies (Netto et al., 2020^[26]).
- **Improved environmental performance:** the modern boiler emits fewer pollutants and supports the company's sustainability goals.

Impact:

- **Advancement in sustainable agroindustry:** this project highlights the role of energy-efficient technologies in enhancing **competitiveness and sustainability** in the agro-processing sector.
- **Risk mitigation through ESI:** The use of energy savings insurance gave Bayas del Sur **confidence in the projected benefits**, enabling the company to secure financing with reduced risk.
- **Scalable success for industrial applications:** the successful outcome of this project positions it as a **scalable model** for similar facilities in Chile and the wider Latin American region looking to upgrade core energy systems (Netto et al., 2020^[26]).

Case study A7: Energy Savings Insurance (ESI) Programme - Morocco

Country: Morocco

Programme: ESI Programme

Partners: BASE (Basel Agency for Sustainable Energy), Moroccan Agency for Energy Efficiency (AMEE), local financial institutions

Target audience: Small and Medium-sized Enterprises (SMEs) across priority sectors

Background: In Morocco, SMEs form the backbone of the economy, yet they often face significant barriers when it comes to adopting energy-efficient technologies. These include limited technical know-how, financing challenges and perceived investment risks. To address these barriers and promote sustainable development, Morocco launched the **Energy Savings Insurance (ESI) Program**, an innovative initiative designed to de-risk energy efficiency investments.

Led by **BASE** in collaboration with **AMEE** and local financial institutions, the ESI Program empowers Moroccan SMEs to transition toward cleaner, more efficient operations while also boosting confidence among investors and insurers through standardised procedures and guaranteed performance.

Project scope and technologies:

The Moroccan ESI Program supports a wide range of **energy-efficient technologies**, including:

- High-efficiency **lighting systems**
- Industrial and commercial **motors**
- **Compressed air systems**
- **Boilers** and heat recovery systems
- **Refrigeration units**
- **Heating and air conditioning (HVAC)**
- **Solar panels** for electricity generation
- **Solar water heaters** for thermal applications

Financial impact:

- **Initial investment:** approximately **USD 31 million** invested in energy efficiency projects during the first **three years** of implementation
- **Total accumulated investment:** reached **USD 141 million** by the end of **six years**
- **Energy savings achieved:** estimated total energy savings of **1 450 GWh** over a **15-year operational period** - equivalent to the annual consumption of thousands of households
- **Cost Savings:** While exact monetary savings are not disclosed, the energy savings imply substantial operational cost reductions and improved financial resilience for participating SMEs (BASE, 2022^[27]).

Key implementation strategies:

The program strategically concentrated on **high-energy-consumption sectors**, such as:

- **Industry** (e.g. food processing, textiles, chemicals)
- **Commercial buildings** (e.g. hotels, retail, office complexes)
- **Agriculture** (e.g. irrigation systems, cold storage)

Impact:

- **Scalable model:** Morocco's experience showed that ESI can be successfully scaled when combined with strong institutional partnerships and local market engagement.
- **Market development:** by lowering perceived risk and increasing technical capacity, the program **stimulated a new market** for energy efficiency solutions tailored to SMEs.
- **Financial innovation:** the use of insurance as a risk-mitigation tool enabled greater access to financing from local banks and investors, even for SMEs with limited collateral or credit history (BASE, 2022^[27]).

Case study A8: ESI Programme - Mongolia

Location: Ulaanbaatar, Mongolia

Programme: Energy Savings Insurance (ESI) Programme

Partners: BASE (Basel Agency for Sustainable Energy), Tenger Insurance, Monhorus

Background: Located in Ulaanbaatar, the capital of Mongolia - one of the world's most polluted and coldest capital cities - **XacBank** recognized the urgent need to address its growing energy demands and environmental impact. The bank's headquarters, which houses an energy-intensive server room, consumed significant amounts of electricity annually. In line with its corporate sustainability goals, XacBank became the **first institution in Mongolia to pilot the Energy Savings Insurance (ESI) model**, aiming to reduce operational costs and greenhouse gas emissions through the implementation of a renewable energy solution.

Project description:

- **Technology implemented:** The project involved the installation of a **rooftop photovoltaic (solar PV) system** with a total capacity of **19.62 kW**. The system was specifically designed for **optimal solar exposure** on the bank's central office building and to support critical energy loads such as the server infrastructure.
- **Energy source transition:** by shifting a portion of its energy use from the grid to solar power, XacBank decreased its dependence on conventional electricity sources, contributing to Mongolia's broader efforts to adopt cleaner, low-emission energy technologies.
- **Total investment:** an investment of approximately **USD 33 000** was made to cover system design, installation and integration into the building's energy supply (BASE, 2024^[25]).

Achieved benefits:

- **Annual energy generation:** the solar PV system generates approximately 22 000 kWh per year, covering about 20% of the building's total electricity needs.
- **Cost savings and efficiency:** by generating its own clean energy, the bank achieved considerable long-term savings on electricity costs while reducing operational risks associated with energy price volatility (BASE, 2024^[25]).
- **Performance assurance:** with support from the Energy Savings Insurance, provided by Tenger Insurance, the bank was assured of the system's energy output - reducing investment risk and strengthening internal confidence in the project's returns.

Impact:

- **Sustainability leadership:** XacBank demonstrated corporate leadership in environmental sustainability, becoming the **first organisation in Mongolia to implement the ESI model** and setting a benchmark for similar institutions in the region.

- **Support for the local green economy:** the project engaged local providers, such as **Monhorus** for installation and engineering services, fostering the growth of Mongolia's green technology sector.
- **Replication potential:** following the project's success, a **second ESI-backed solar project** was announced in **December 2023** for an **E-Mart supermarket**, with a **32-kW capacity system** expected to generate **80 000 kWh annually**. This highlights the replicability of the model across sectors (BASE, 2024^[25]).

B. Energy efficiency insurance

Five case studies are presented below on the application of the energy efficiency insurance in the United Kingdom (UK) and United States of America (USA).

Case study B1: Energy efficiency insurance in a 5-Star hotel - United Kingdom

Location: United Kingdom

Programme: HSB Energy Efficiency Insurance (Munich Re)

Partners: Munich Re/HSB

Background: an independent 5-star hotel in the UK, built in 1996 and comprising 174 rooms, faced high operational costs due to inefficient energy use, with an **annual energy spend of approximately USD 4 million**. In an effort to modernise its systems, reduce emissions and improve profitability, the hotel committed to a multi-technology energy efficiency retrofit. Leveraging the **HSB Energy Efficiency Insurance** solution provided by Munich Re, the hotel was able to access financial protection for both performance and operational risks.

Project description:

- Technology implemented: A total of seven energy-saving initiatives, including:
- Combined Heat and Power (CHP) plant
- LED lighting installation
- Building Management System (BMS) upgrade
- Variable Speed Drives (VSDs)
- Rooftop solar PV panels
- Upgrades to air conditioning and extraction systems
- **Building Type:** Independent 5-star hotel (174 rooms)
- **Total Investment (CapEx):** **USD 3,500,000** (HSB UK and Ireland, 2026^[28])

Achieved benefits:

- **Estimated annual energy savings:** approximately **USD 575 000 per year**, reducing the hotel's energy intensity and operational costs.
- **Annual revenue from energy initiatives:** around **USD 375 000**, showcasing clear commercial gains.
- **Insurance coverage:**
 - **Material damage and breakdown:** **USD 3 500 000**
 - **Loss of revenue:** **USD 375 000**

- **Insured energy savings: USD 575 000** (over a 5-year term) (HSB UK and Ireland, 2026^[28])

Insurance solution: The HSB Energy Efficiency Insurance offered by Munich Re provided a **comprehensive risk mitigation framework**, including:

- Coverage for equipment breakdown, material damage and underperformance of energy-saving systems
- Protection against shortfalls in guaranteed savings
- Support for business interruption losses related to energy systems

Impact:

- **De-risked investment:** The insurance made it easier for the hotel to **secure project financing**, giving confidence to lenders and investors by protecting against technical and financial uncertainties.
- **Sector-specific application:** demonstrated how **customised insurance packages** for hospitality can unlock large-scale energy efficiency upgrades.
- **Financial and environmental payoff:** the project showcases the **synergy of technical solutions and financial instruments**, combining infrastructure improvement with guaranteed returns.

Case study B2: Munich Re's Performance Guarantee Insurance for Xicato's LED modules

Client: Xicato, a U.S.-based LED lighting company

Partner: Munich Re

Traditional LED warranties only cover failures or workmanship defects for one year. Customers, especially in retail and hospitality, demanded stronger assurances on long-term performance, specifically colour consistency and brightness.

Solution:

Xicato introduced a 5-year performance guarantee on its LED modules:

- Color stability: Minimal deviation over time.
- Lumen maintenance: Minimum 70% light output after five years.
- Munich Re underwrote this guarantee, a portion of the financial risk after auditing Xicato's development and manufacturing processes (Programbusiness.com, 2013^[29]).

Industry significance:

This coverage complements Munich Re's portfolio of performance guarantees for renewable technologies, advancing insurance's role in accelerating adoption of sustainable innovations.

Impact:

- First of its kind: pioneering insurance-backed LED performance guarantee.
- Xicato reduced long-term risk exposure on its balance sheet, freeing capital for growth.
- Provided independent validation and boosted buyer confidence.

Key learnings:

- Insurance-backed guarantees reassure customers about long-term LED performance (colour stability, brightness).

- Risk transfer to Munich Re helped Xicato reduce financial liability and invest in growth.
- The 5-year insured warranty differentiated Xicato from competitors with standard 1-year coverage (Programbusiness.com, 2013^[29]).

Case study B3: Energy Efficiency Insurance in a hospital facility - United States

Location: United States

Program: HSB Energy Efficiency Insurance

Partners: HSB (Munich Re), Energy Services Company (ESCO)

Background: the hospital facility, originally built in 1970, spans 580 000 square feet. With an annual energy cost baseline of USD 430 000, the hospital recognised the need for significant energy upgrades to reduce costs, enhance operational efficiency and meet sustainability goals. The hospital engaged with an Energy Services Company (ESCO) for an energy efficiency retrofit, which included comprehensive upgrades to the building's infrastructure (Hemic, 2016^[30]).

Project description:

- **Technology implemented:** the project included 12 energy conservation measures, which were designed to reduce energy usage and improve operational efficiency. These measures included:
- **HVAC upgrades:**
 - Lighting replacements with energy-efficient LEDs
 - Installation of variable frequency drives (VFDs) for motors
 - Water conservation measures (such as low-flow fixtures and systems) (Hemic, 2016^[30])

Building type: hospital facility (multi-story, 580 000 sq. ft.)

Total investment (CapEx): USD 2.4 million

Achieved benefits:

- **Annual energy savings:** the project was projected to achieve USD 260 000 in annual energy savings.
- **Insured energy savings:** HSB's energy efficiency insurance covered 90% of the guaranteed annual energy savings, which amounts to USD 234 000 per year (Hemic, 2016^[30]).
- **Insurance coverage:** the policy covered shortfalls in annual aggregate efficiency savings as a result of deficiencies in design, implementation and/or performance. Losses were verified via annual measurement and verification reports.

Insurance and loan term: the project was financed with a 10-year loan.

- **Risk mitigation:** the energy efficiency insurance covered risks related to the shortfall in energy savings and provided coverage against losses. Additionally, the credit enhancement provided by HSB's solution allowed the ESCO to show significant improvement in the credit rating of the project, which helped reduce the overall cost of the project loan.
- **Financial outcome:** With the insurance coverage in place, the hospital was able to reduce financial risks associated with the project, making the large-scale energy improvements more feasible.

Impact:

- **Risk reduction for project stakeholders:** the energy efficiency insurance mitigated financial risks, protecting the project owner, contractors and stakeholders from potential performance shortfalls and failures to meet energy savings targets.
- **Improved project financing:** the credit enhancement feature helped improve the credit rating of the project, which reduced the overall financing cost and made the investment in energy-saving technologies more financially attractive.

Operational efficiency: with the annual energy savings and improved systems in place, the hospital significantly reduced operational costs while improving energy efficiency across its facility, aligning with sustainability and financial goals.

Case study B4: Performance Insurance for fuel cell technology

Background: in 2013, XL Group's Complex Accounts unit, Munich Re's Green Tech Solutions team, and New Energy Risk (NER) collaborated to offer an innovative performance insurance policy for a portfolio of fuel cell servers installed across the U.S.

Challenge: fuel cell technology manufacturers faced challenges in meeting performance expectations, which could hinder financing and investor confidence.

Solution: The insurance cover provided critical support for project financing by insuring the fuel cells' performance over 15 years, ensuring reliability even if manufacturers couldn't meet warranty obligations. The USD 99 million bond issued to finance the portfolio received an investment-grade rating, enhancing its attractiveness to investors (New Energy Risk, 2014^[31]).

Outcome: this collaboration marked the first time that insurance coverage provided by XL Group and Munich Re contributed to a financing bond receiving an investment-grade rating, broadening its appeal to a wider range of investors. The initiative demonstrated the importance of insurance in making clean technology projects more attractive and financially viable (New Energy Risk, 2014^[31]).